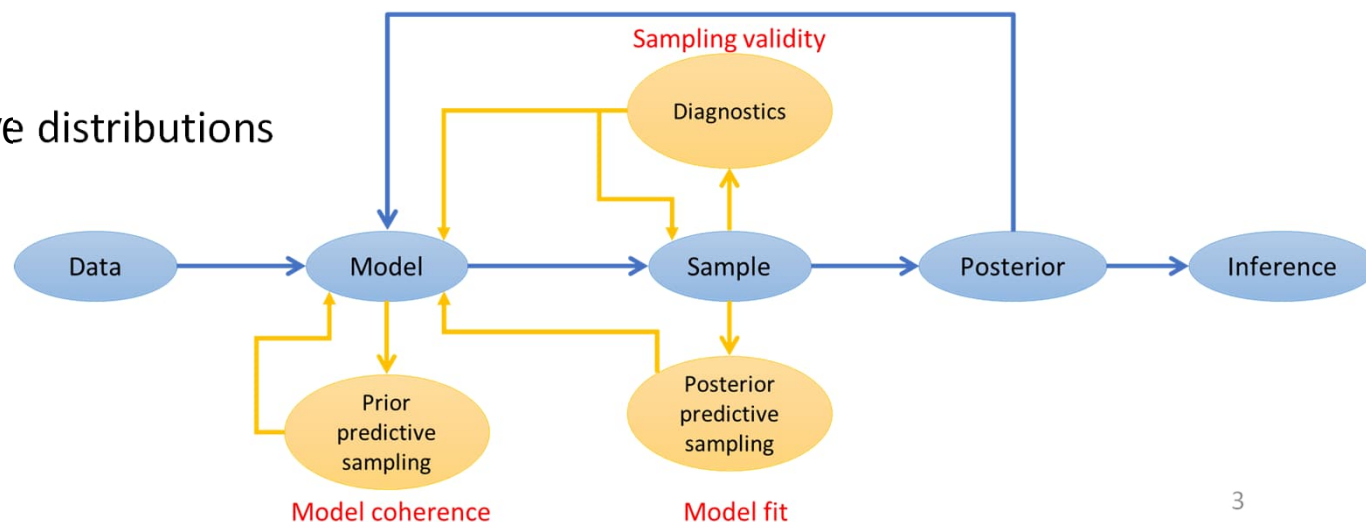
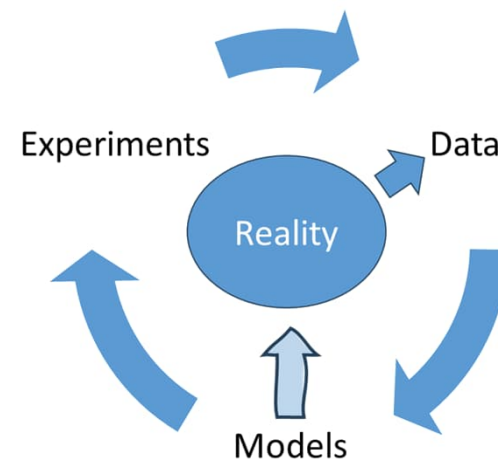


Statistics
367-1-4361
Hierarchical Models
Opher Donchin

Key concepts so far

- Data
 - Coin flips
- Model
 - Binomial likelihood with beta prior
- Posterior distribution
 - Sampled by PyMC
 - Using NUTS
- Workflow
 - Prior and posterior predictive distributions
 - Sampling diagnostics



Today's lecture

- So far data has been very simple
 - Flipping a coin
- But really data is complex
 - What can we do?

5.1 Grouped data

In most experiments, data is structured

Grouped data is common

- Multiple drugs applied to different cell cultures
- Each subject does multiple sessions of EEG
 - And each session has multiple trials
- Animals are divided into treatment groups
 - And each animal is measured at multiple time points
- Each subject tries multiple prostheses
 - Each prosthetic is evaluated over several days
 - On every day there are several trials

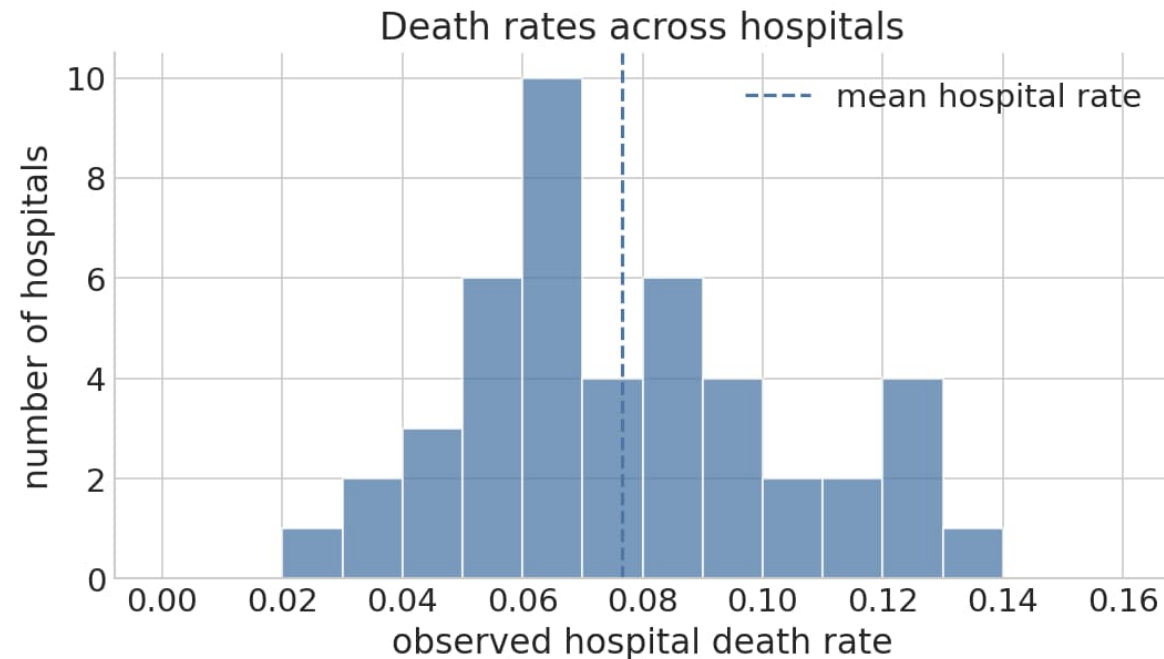
Death rate following cardiac arrest

- Hospital death rates can differ between:
 - Hospitals
 - Boroughs
 - Equipment available

	Hospital	Borough	Type	Cases	Deaths	DeathRate
0	Bronx-Lebanon Hospital Center - Concourse Divi...	The Bronx	PCI	209	20	0.095694
1	Jacobi Medical Center	The Bronx	Non-PCI	92	5	0.054348
2	Lincoln Medical & Mental Health Center	The Bronx	Non-PCI	133	4	0.030075
3	Montefiore Med Center - Jack D Weiler Hospital...	The Bronx	PCI	336	18	0.053571
4	Montefiore Medical Center - Henry & Lucy Moses...	The Bronx	PCI	267	24	0.089888

Do hospitals have different death rates?

- Differences seem large
- But maybe they are just noise

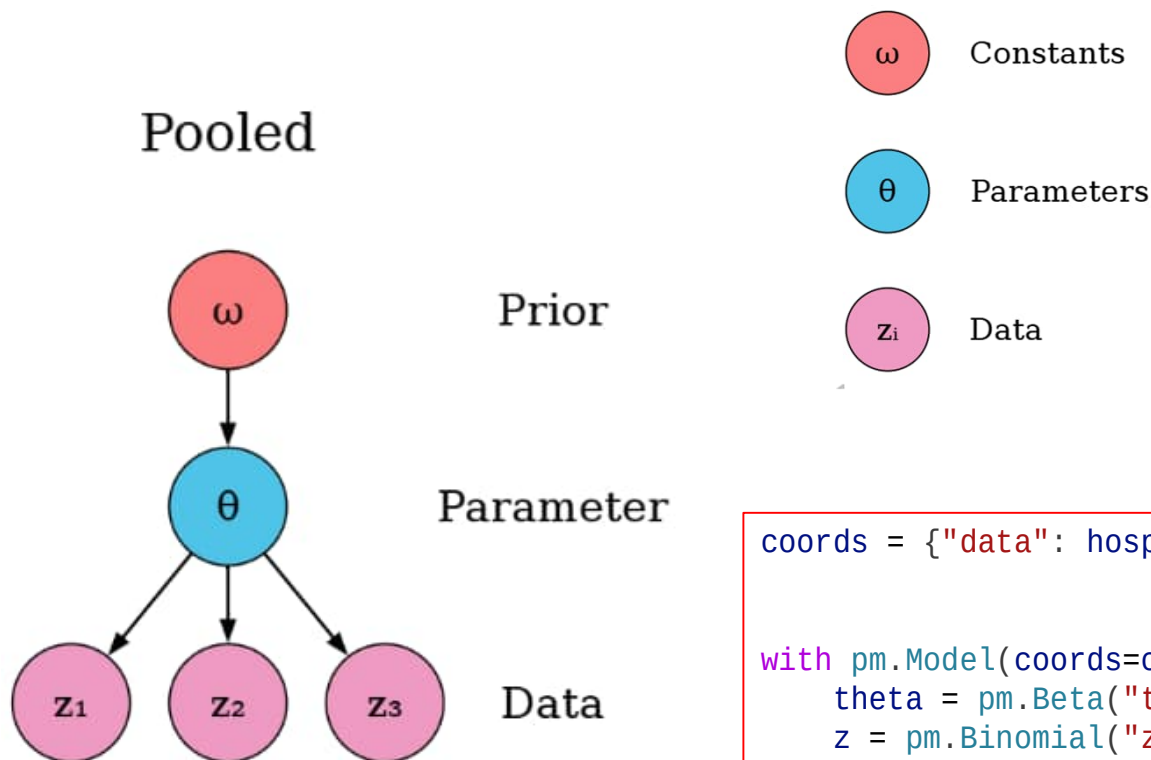


5.2 Pooled model

A pooled model ignores the structure in the data

We can model the death rate across hospitals

- Total death rate: 698 deaths in 9,235 cases: 7.56%
- Taken across hospitals
 - Call this a pooled model



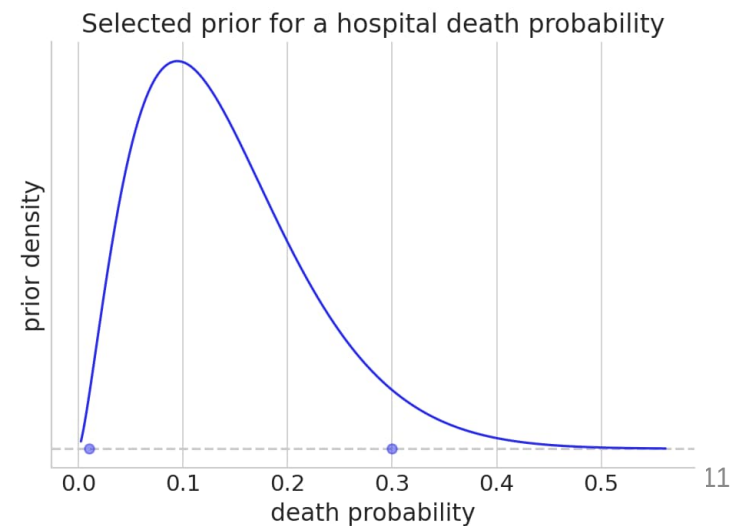
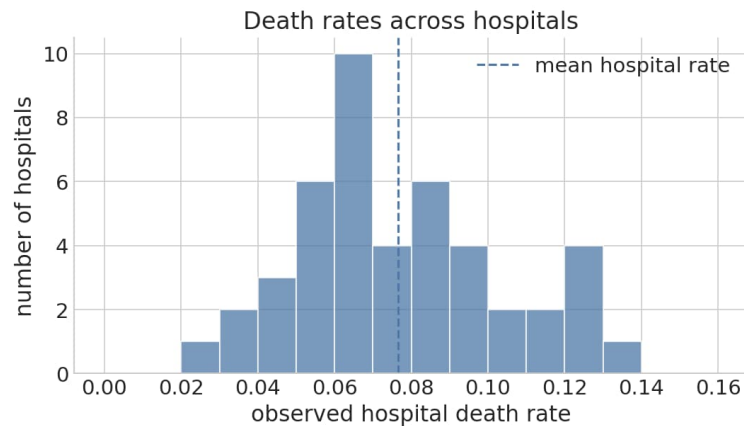
```
coords = {"data": hospital_names}
```

```
with pm.Model(coords=coords) as pooled_binomial_model:  
    theta = pm.Beta("theta", alpha=omega_alpha, beta=omega_beta)  
    z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")
```

We can use a weakly informed prior

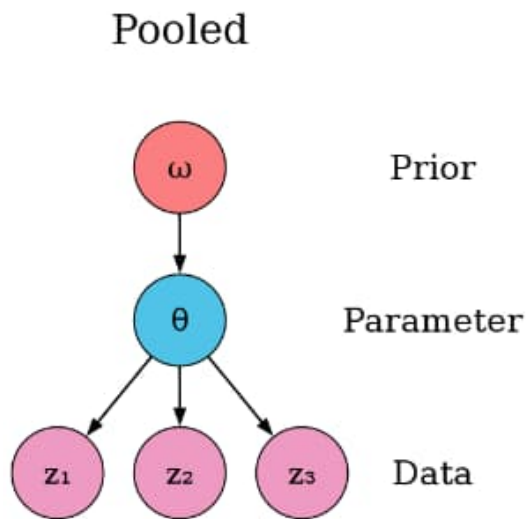
- In a realistic scenario, we would know if most cardiac arrests result in deaths
- Here, we use the data as a guide

```
death_rate_prior, ax = pz.maxent(  
    pz.Beta(), lower=0.01, upper=0.30,  
    mass=0.95, plot=True,  
)  
omega_alpha = float(death_rate_prior.alpha)  
omega_beta = float(death_rate_prior.beta)
```



We can model the death rate across hospitals

- Total death rate: 698 deaths in 9,235 cases: 7.56%
- Taken across hospitals
 - Call this a pooled model



```
coords = {"data": hospital_names}

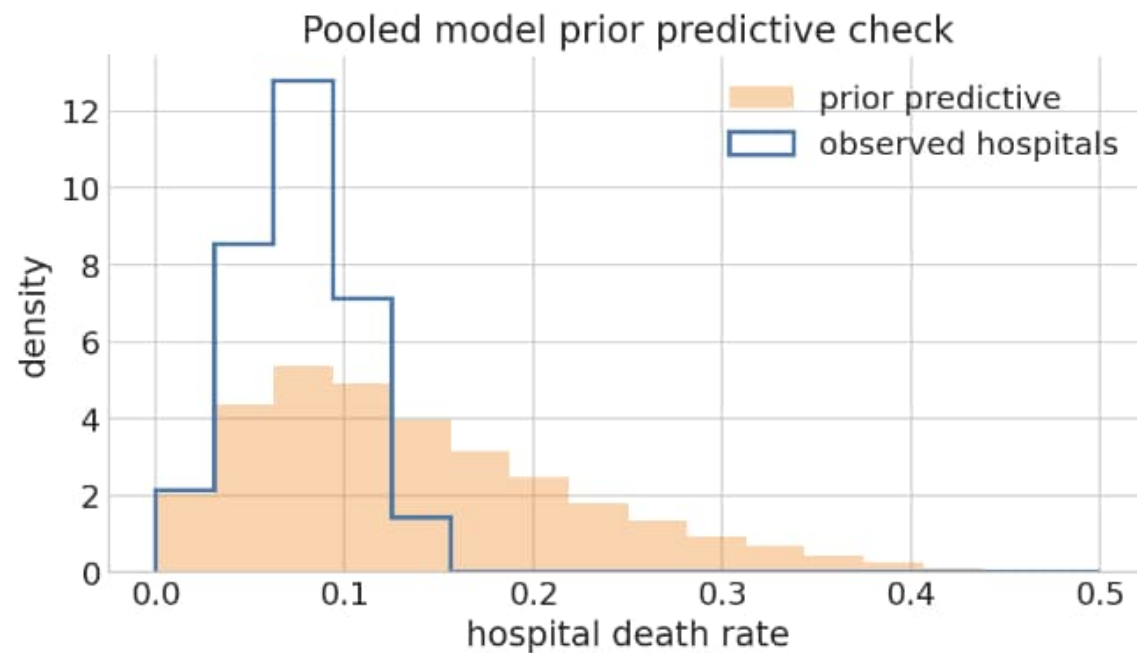
with pm.Model(coords=coords) as pooled_binomial_model:
    theta = pm.Beta("theta", alpha=omega_alpha, beta=omega_beta)
    z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")
```

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.076	0.003	0.071	0.081	0.0	0.0	1235.0	2392.0	1.01

Check prior predictive for plausability

- Prior predictive data should be sane
- Prior predictive should be broader than data

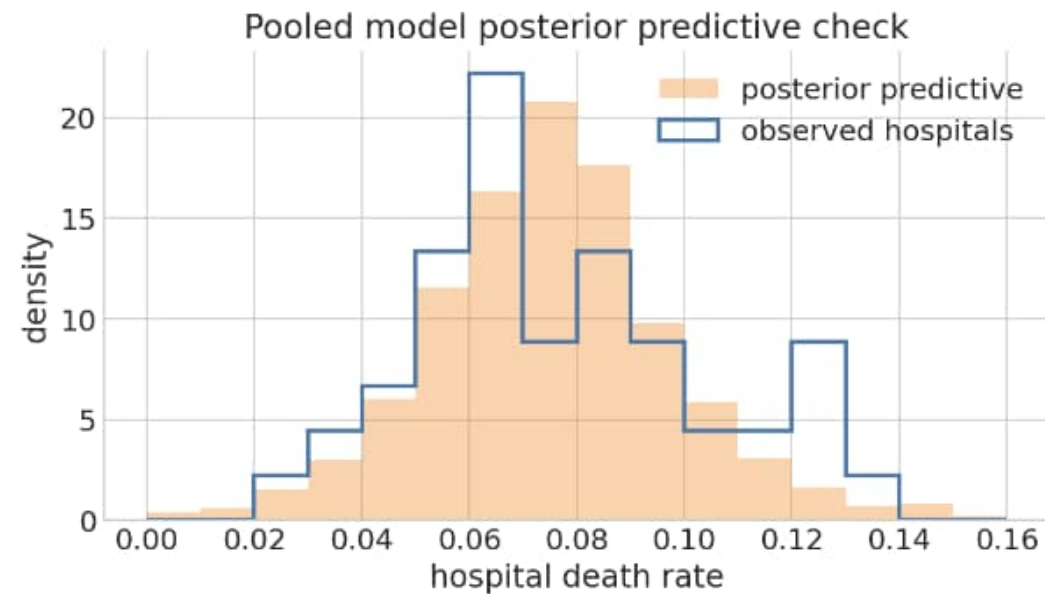
```
with pooled_binomial_model:  
    idata_pooled_prior = pm.sample_prior_predictive(  
        samples=1000, random_seed=124,  
    )
```



Check posterior predictive for model fit

- Real hospitals seem more variable than posterior predictive hospitals

```
with pooled_binomial_model:  
    idata_pooled_posterior = pm.sample_posterior_predictive(  
        idata_pooled, random_seed=125,  
    )
```

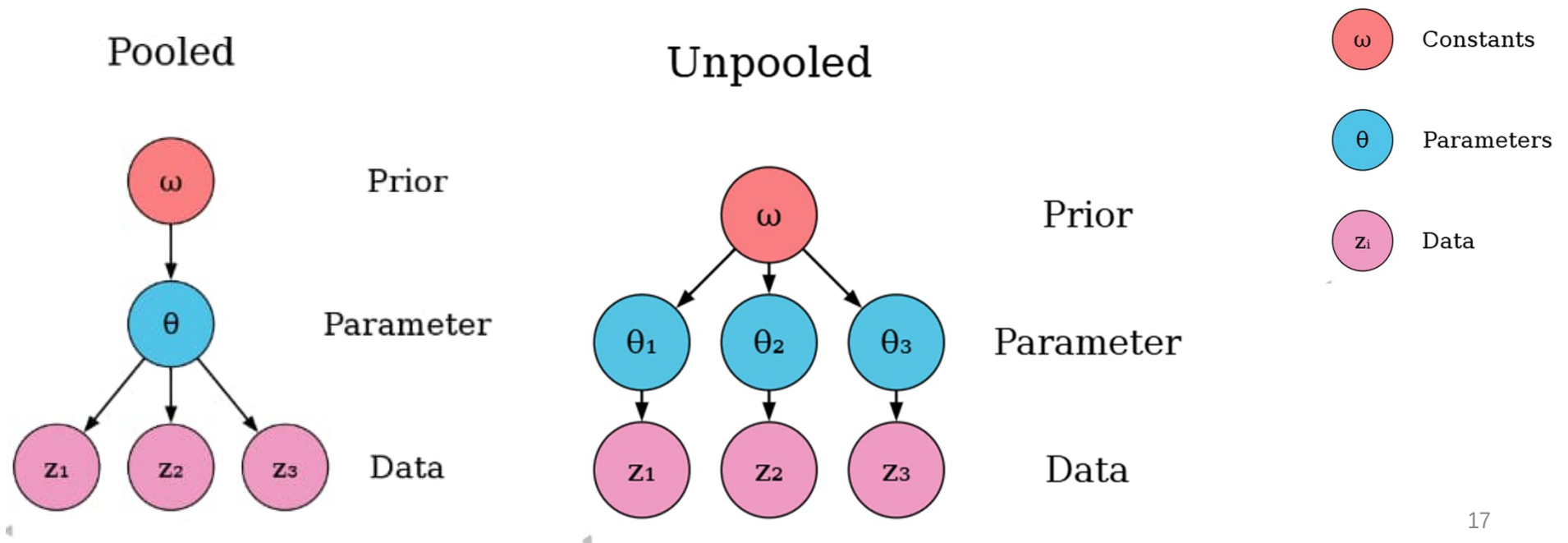


5.3 Unpooled model

Unpooled models compartmentalize the different parts

Unpooled model: separate probabilities

- Data from each hospital is fit with a different parameter



The unpooled model uses multiple theta parameters

- Specify that with a dims parameter on the theta parameter

Pooled

```
with pm.Model(coords=coords) as pooled_binomial_model:  
    theta = pm.Beta("theta", alpha=omega_alpha, beta=omega_beta)  
    z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")
```

Unpooled

```
with pm.Model(coords=coords) as unpooled_binomial_model:  
    theta = pm.Beta("theta", alpha=omega_alpha, beta=omega_beta, dims="data")  
    z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")
```


Each theta parameter has a different value

- Much larger uncertainty in individual thetas

Pooled

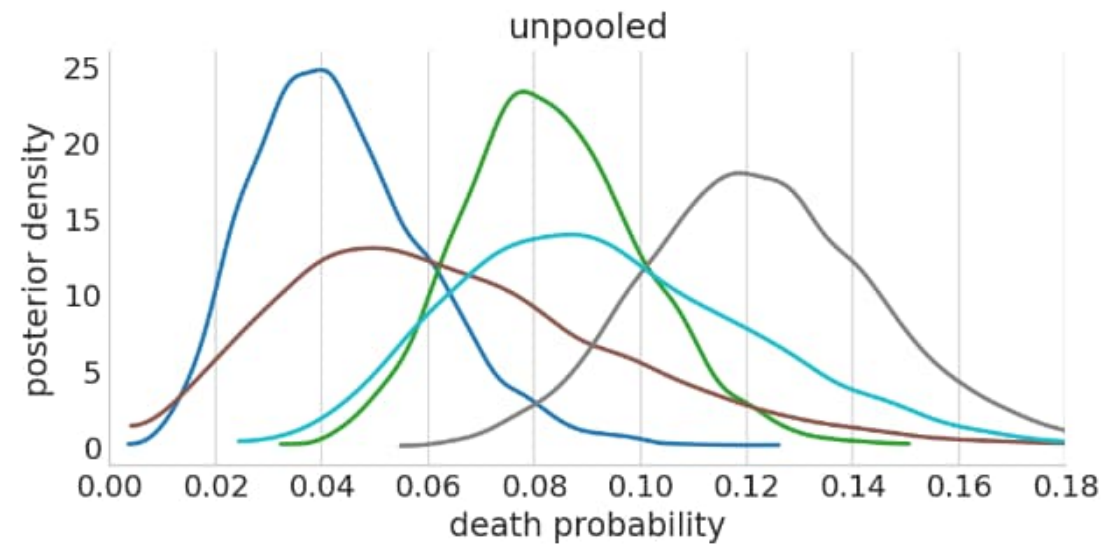
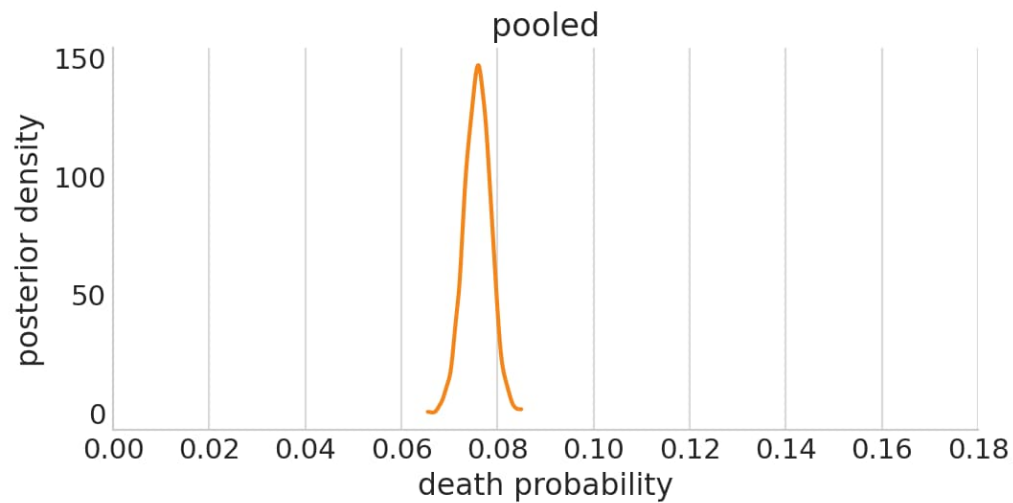
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta	0.076	0.003	0.071	0.081	0.0	0.0	1235.0	2392.0	1.01

Unpooled

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
theta[Bellevue Center]	0.044	0.017	0.014	0.075	0.0	0.000	4308.0	2655.0	1.0
theta[Lenox Hill]	0.083	0.017	0.050	0.115	0.0	0.000	5627.0	2932.0	1.0
theta[Harlem Center]	0.065	0.033	0.013	0.126	0.0	0.001	3974.0	2424.0	1.0
theta[Forest Hills]	0.123	0.022	0.082	0.167	0.0	0.000	5764.0	2549.0	1.0
theta[Metropolitan Center]	0.094	0.029	0.045	0.149	0.0	0.000	5653.0	2728.0	1.0

Pooled and unpooled posterior plots

- Posterior plots show the increased variability
 - No information is shared across the hospitals



5.4 Hierarchical model

Hierarchical models model the structure

Rewriting our models

- Colors for type of variable
- Add the implicit parameters of the prior

Constant
Parameter
Data

$$\theta_g \sim p(\omega) \text{ for } g \in [1 \dots 4]$$

$$y_{g,i} \sim p(\theta_g)$$

$$p(\{\theta_g\} | \{y_i\}) \propto p(\{y_i\} | \{\theta_g\}) p(\{\theta_g\} | \omega)$$

Make ω a parameter

Constant
Parameter
Data

- Factor the prior: definition of conditional probability
- Also y doesn't depend on ω : depends only on θ

$$\omega \sim p(\eta)$$

$$\theta_g \sim p(\omega) \text{ for } g \in [1 \dots 4]$$

$$y_{g,i} \sim p(\theta_g)$$

$$\begin{aligned} p(\{\theta_g\}, \omega | \{y_i\}) &\propto p(\{y_i\} | \{\theta_g\}, \omega) p(\{\theta_g\}, \omega | \eta) \\ &\propto p(\{y_i\} | \{\theta_g\}, \omega) p(\{\theta_g\} | \omega) p(\omega | \eta) \\ &\propto p(\{y_i\} | \{\theta_g\}) p(\{\theta_g\} | \omega) p(\omega | \eta) \end{aligned}$$

Hyperparameters and hyperpriors

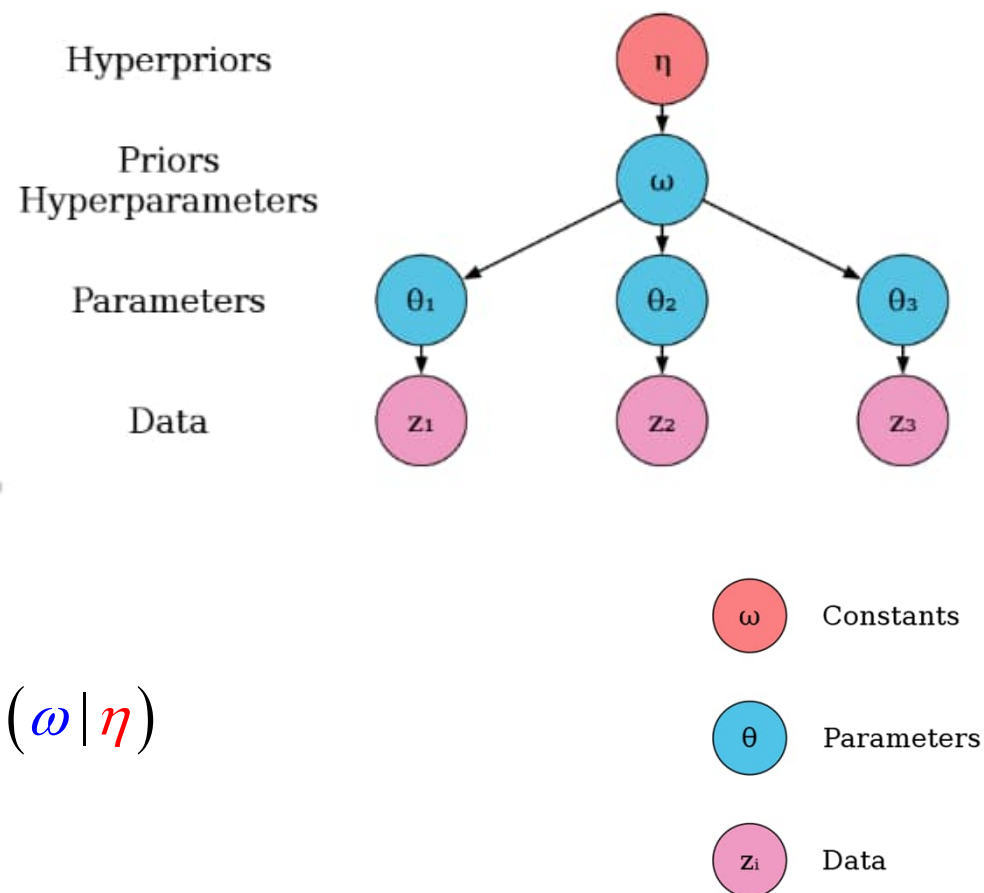
- ω is now called a hyperparameter
- The prior for ω is called a hyperprior
- Its parameters are constants

$$\omega \sim p(\eta)$$

$$\theta_g \sim p(\omega) \text{ for } g \in [1 \dots 4]$$

$$y_{g,i} \sim p(\theta_g)$$

$$p(\{\theta_g\}, \omega | \{y_i\}) \propto p(\{y_i\} | \{\theta_g\}) p(\{\theta_g\} | \omega) p(\omega | \eta)$$



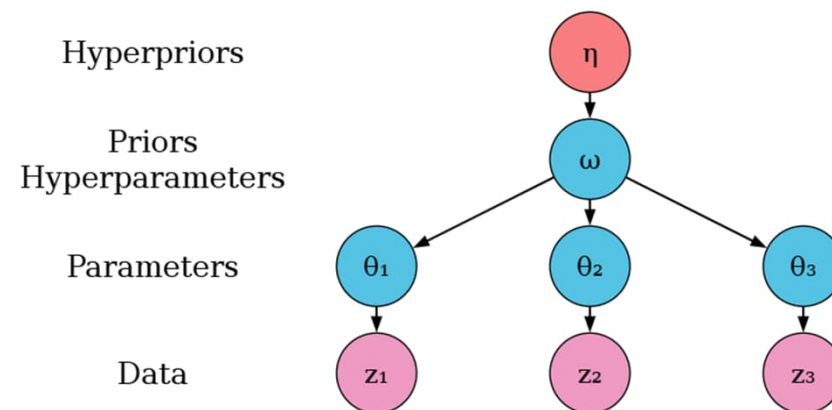
The hierarchical model causes information sharing

$$\omega \sim p(\eta)$$

$$\theta_g \sim p(\omega) \text{ for } g \in [1 \dots 4]$$

$$y_{g,i} \sim p(\theta_g)$$

- Posterior of θ_g depends on y and ω
- Posterior of ω depends on θ_g and $p(\eta)$
- Practically speaking:
 - Each θ_g is now influenced by all θ_g
 - Each θ_g is informed by more data

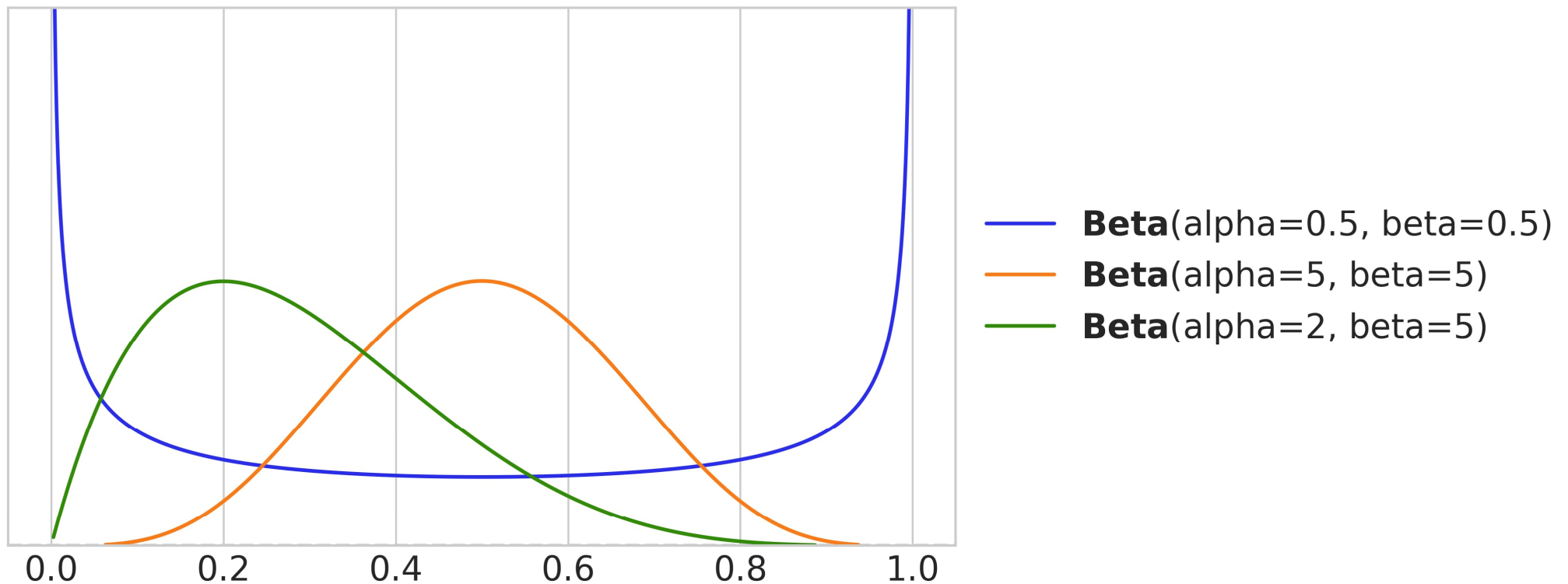


5.4 Parameterizing Beta

The Beta distribution can be parameterized in different ways

Beta distribution parameterized by α and β

- α : number of successes
- β : number of failures

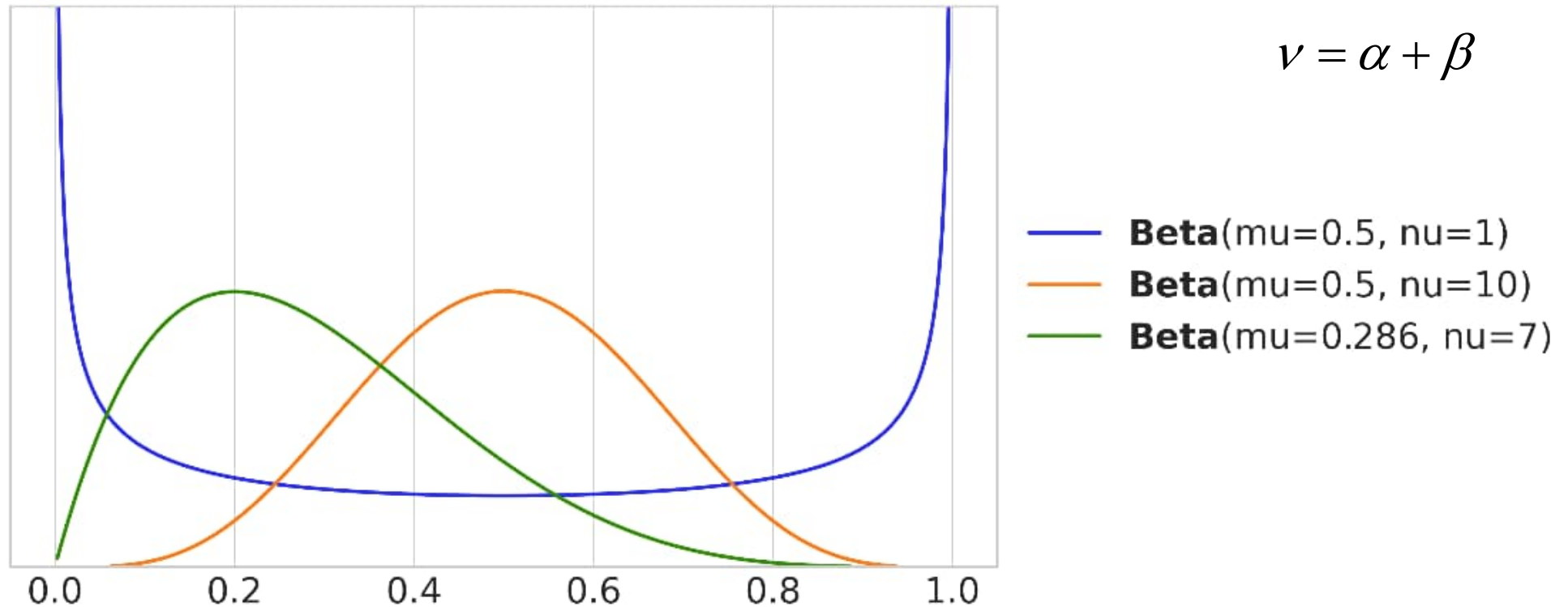


Equivalent parameterization by μ and ν

- μ : mean
- ν : concentration

$$\mu = \frac{\alpha}{\alpha + \beta}$$

$$\nu = \alpha + \beta$$

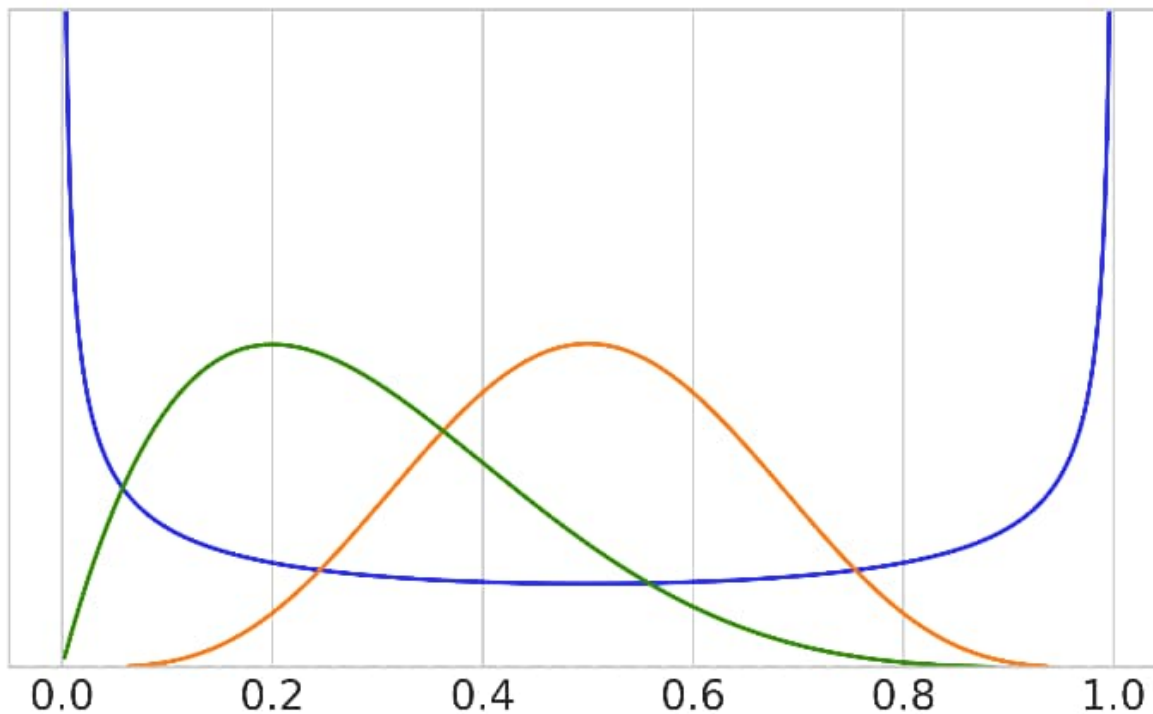


Equivalent parameterization by μ and σ

- μ : mean
- σ : standard deviation

$$\mu = \frac{\alpha}{\alpha + \beta}$$

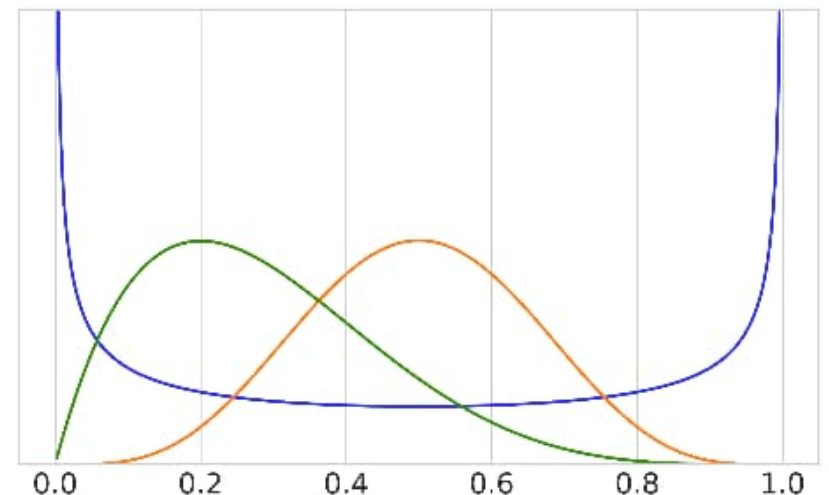
$$\sigma = \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}}$$



- **Beta**(mu=0.5, sigma=0.354)
- **Beta**(mu=0.5, sigma=0.151)
- **Beta**(mu=0.286, sigma=0.16)

Parameterizations of the Beta distribution

- As a prior equivalent to a number of successes and failures
 - α successes, β failures
- As a prior equivalent to number of attempts and mean success rate
 - μ mean, ν number of attempts
- By the mean and the width
 - μ mean, σ standard deviation
- We will use μ, ν

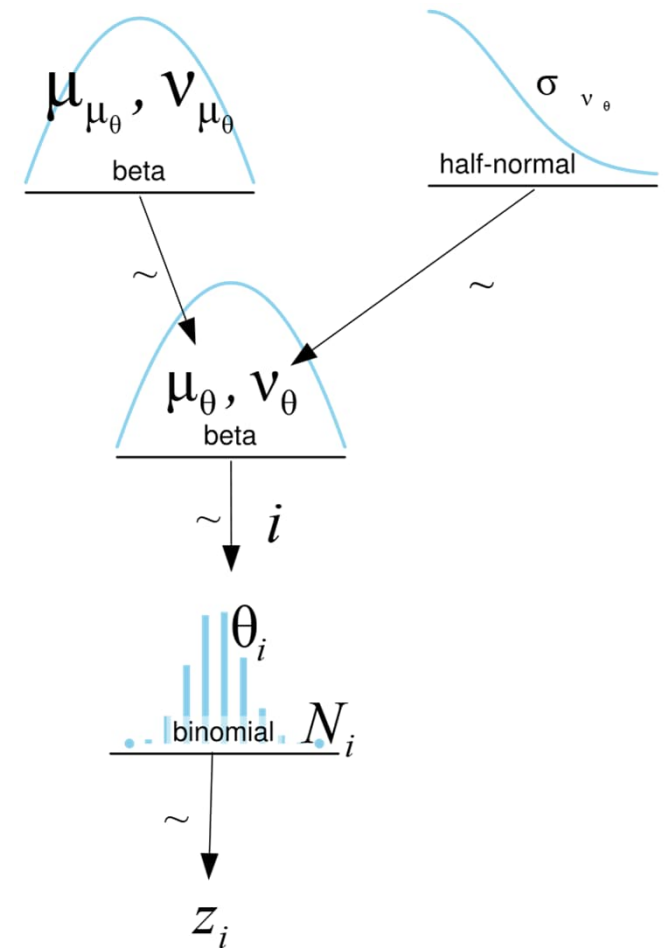


5.5 Coding hierarchical models

Approaches to writing down complex models

Graphical representation shows hierarchical structure

- Data on the bottom
- Next: parameters
- Next: hyper-parameters
- Then: Hyper-priors on top

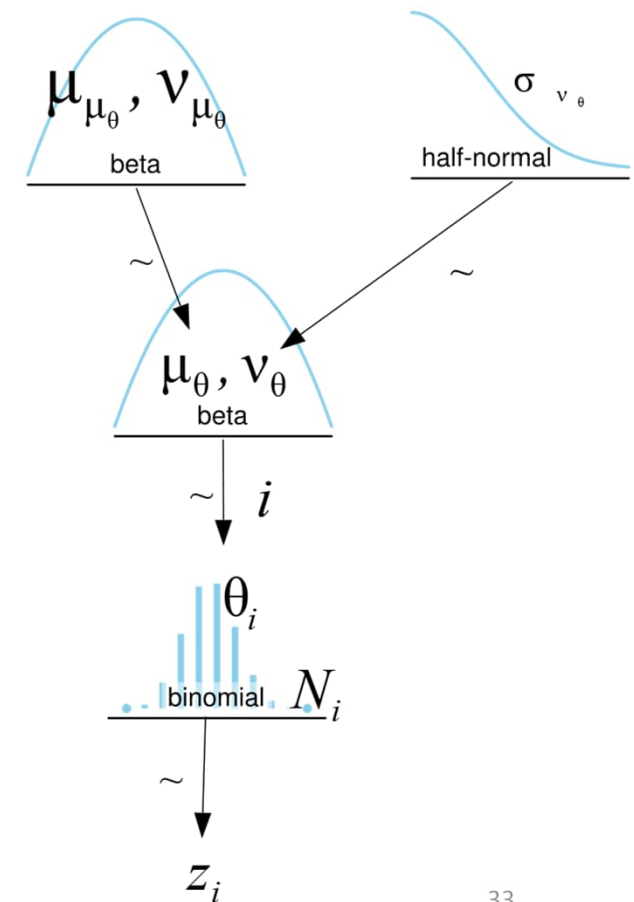


Graphical representation translates to code

- z : Binomially distributed likelihood
 - With parameter θ
- θ : Beta distributed parameter
 - With μ_θ, ν_θ unknown
- μ_θ, ν_θ : Hyperparameters
 - With prior distributions

```
with pm.Model(coords=coords) as hospital_beta_binomial_model:  
    mu_theta = pm.Beta("mu_theta", mu=mu_mu_theta, nu=nu_mu_theta)  
    nu_theta = pm.HalfNormal("nu_theta", sigma=sigma_nu_theta)
```

```
theta = pm.Beta("theta", mu=mu_theta, nu=nu_theta, dims="data")  
z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")
```

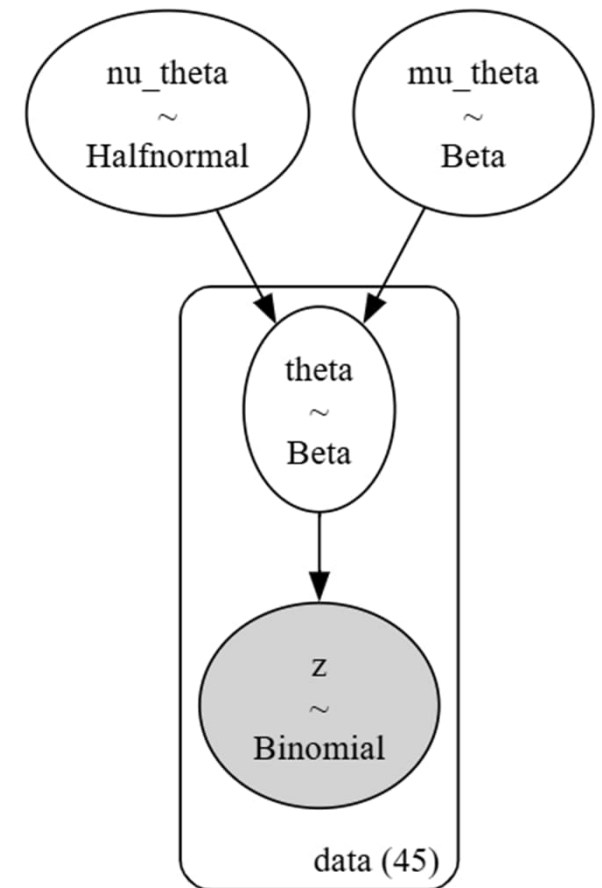


The code can be used to generate a graphical model

```
with pm.Model(coords=coords) as hospital_beta_binomial_model:
    mu_theta = pm.Beta("mu_theta", mu=mu_mu_theta, nu=nu_mu_theta)
    nu_theta = pm.HalfNormal("nu_theta", sigma=sigma_nu_theta)

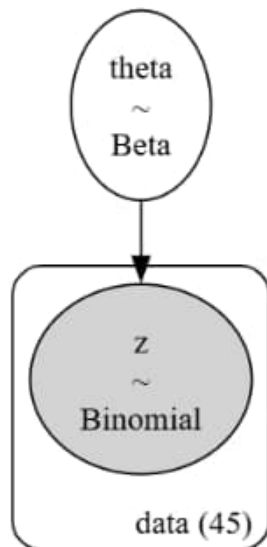
    theta = pm.Beta("theta", mu=mu_theta, nu=nu_theta, dims="data")
    z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")

pm.model_to_graphviz(hospital_beta_binomial_model)
```

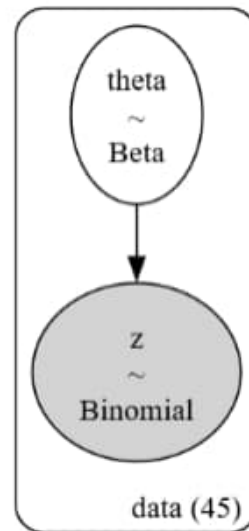


Hierarchical models have an extra layer

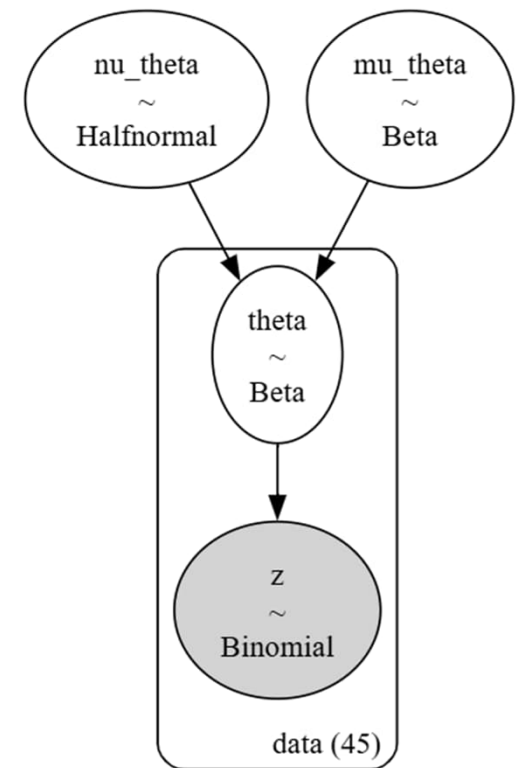
Pooled



Unpooled

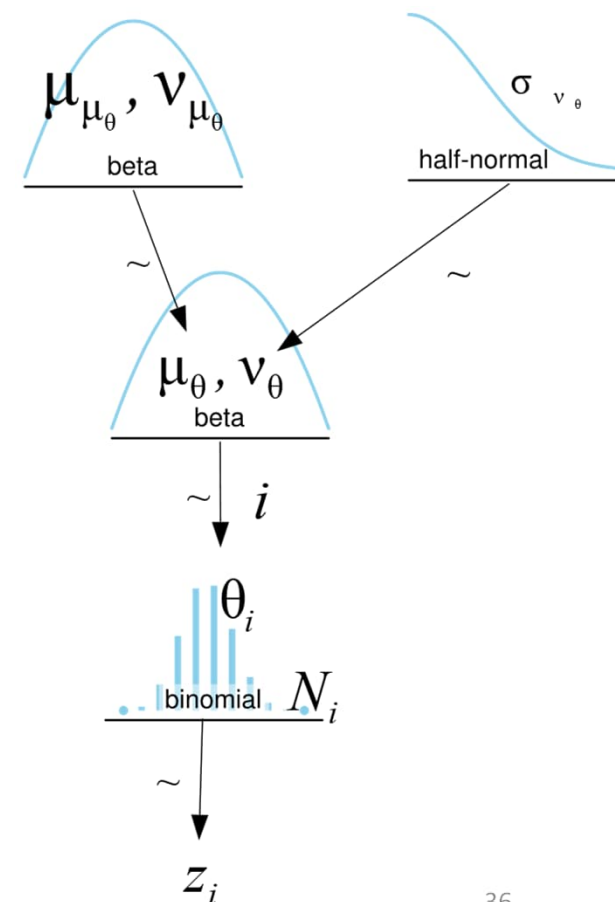


Hierarchical



Choosing hyperpriors

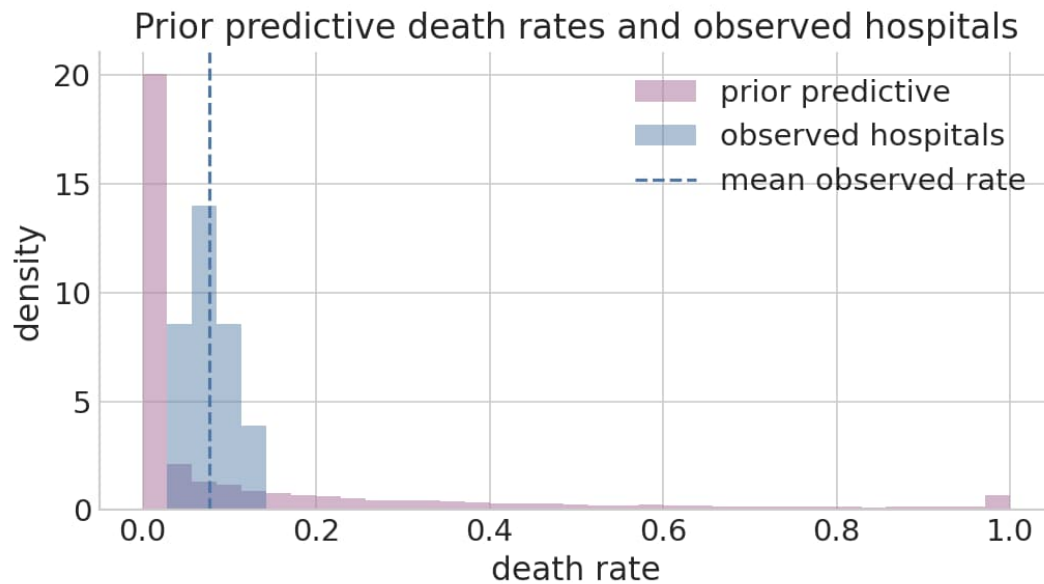
- $\mu_{\mu_{\theta}}, \nu_{\mu_{\theta}}$
 - The beta prior for the overall mean
 - Probably near the middle of the data
 - Set to match ω_{α} and ω_{β}
 - Probably wider than we need
- $\sigma_{\nu_{\theta}}$
 - How far apart can the hospitals be
 - Chosen to be wider than actual hospitals



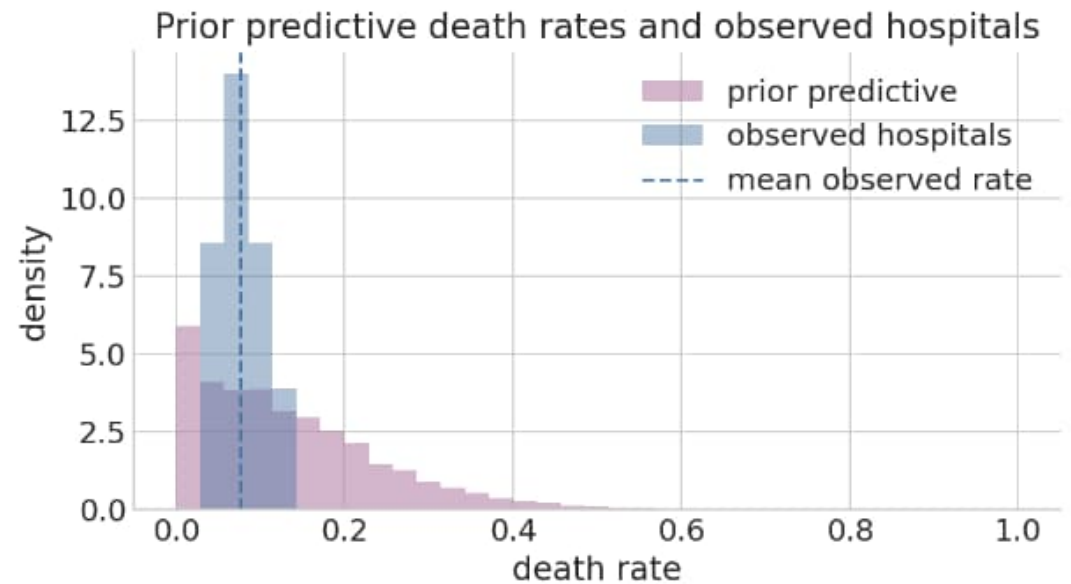
Examine prior predictive

- Initial guess was too wide

$$\sigma_{\theta} = 0.2$$



$$\sigma_{\theta} = 0.05$$



5.6 Fitting hierarchical models

Understanding the samples and using the workflow

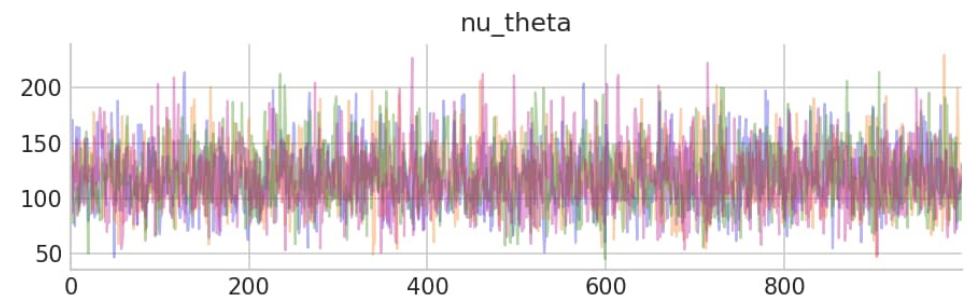
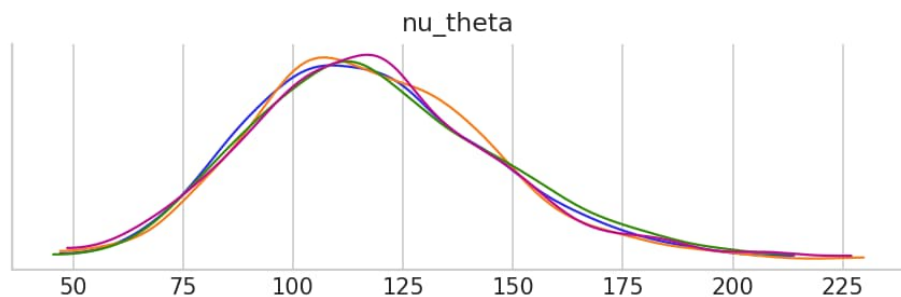
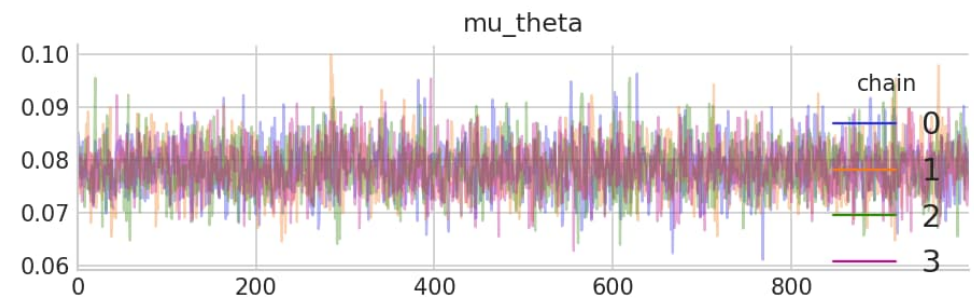
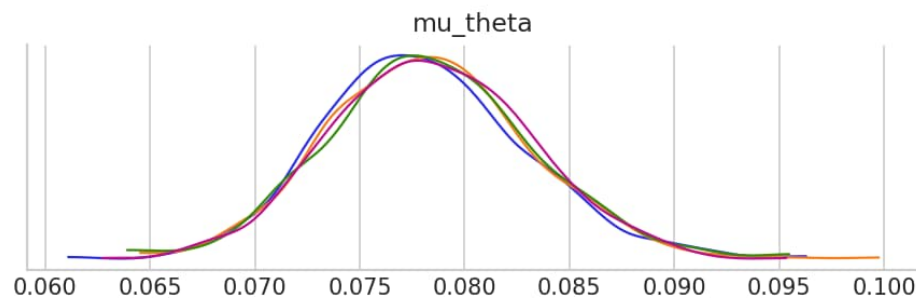
Sampling diagnostics look good

- Remind yourself what r_{hat} , ess , and mcse mean
- Std of parameters also encouraging

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_theta	0.078	0.005	0.069	0.087	0.0	0.000	3602.0	2840.0	1.0
nu_theta	118.077	27.888	67.031	171.123	0.5	0.398	3119.0	3088.0	1.0
theta[Bellevue Center]	0.053	0.014	0.028	0.080	0.0	0.000	4717.0	3091.0	1.0
theta[Lenox Hill]	0.079	0.015	0.051	0.106	0.0	0.000	5121.0	2275.0	1.0
theta[Harlem Center]	0.067	0.020	0.030	0.103	0.0	0.000	5846.0	3192.0	1.0
theta[Forest Hills]	0.106	0.018	0.074	0.140	0.0	0.000	5524.0	3018.0	1.0
theta[Metropolitan Center]	0.080	0.020	0.044	0.117	0.0	0.000	5669.0	3190.0	1.0

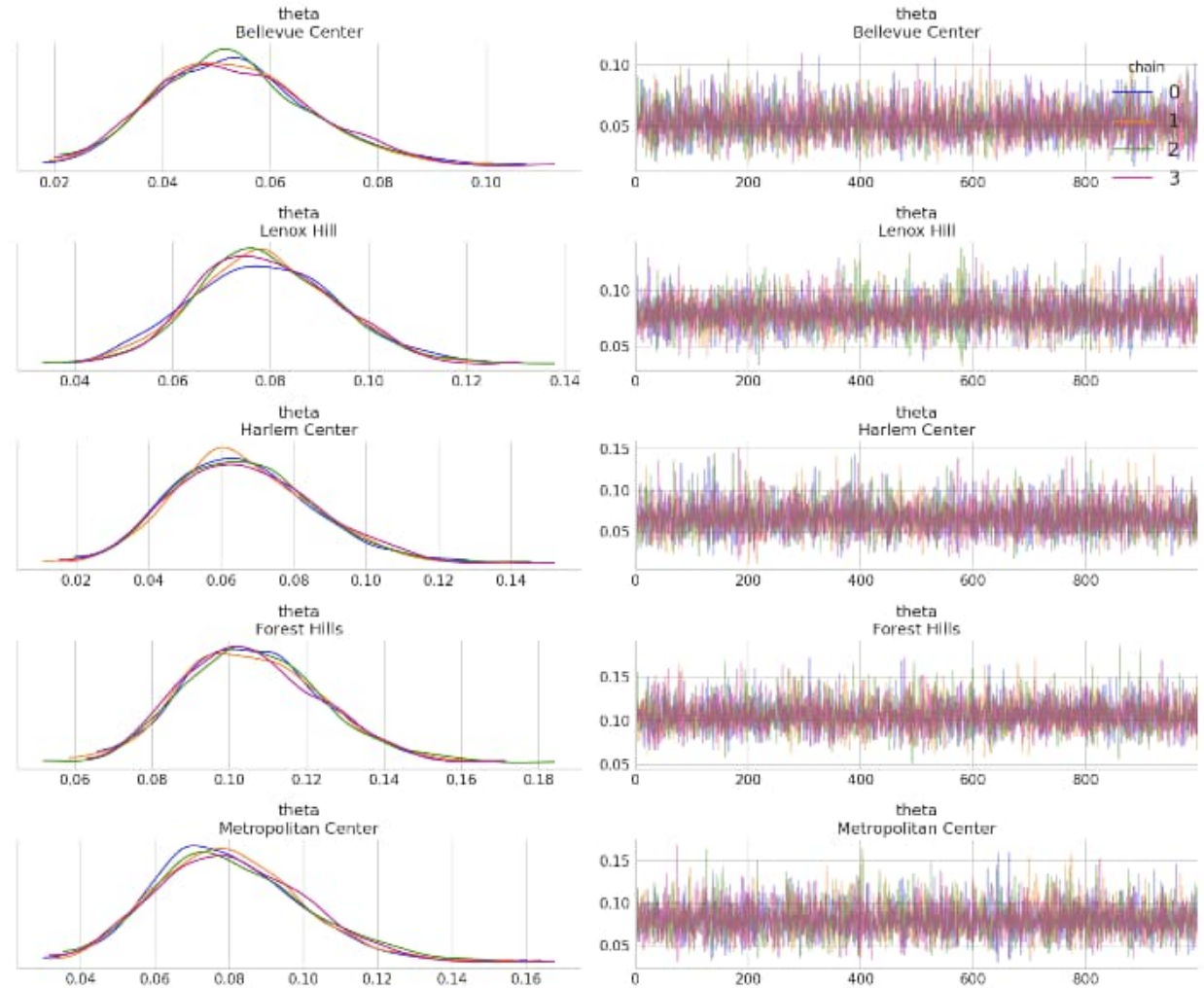
Trace plots for hyperparameters look good

- Remind yourself what $\hat{r}$, ess, and mcse mean



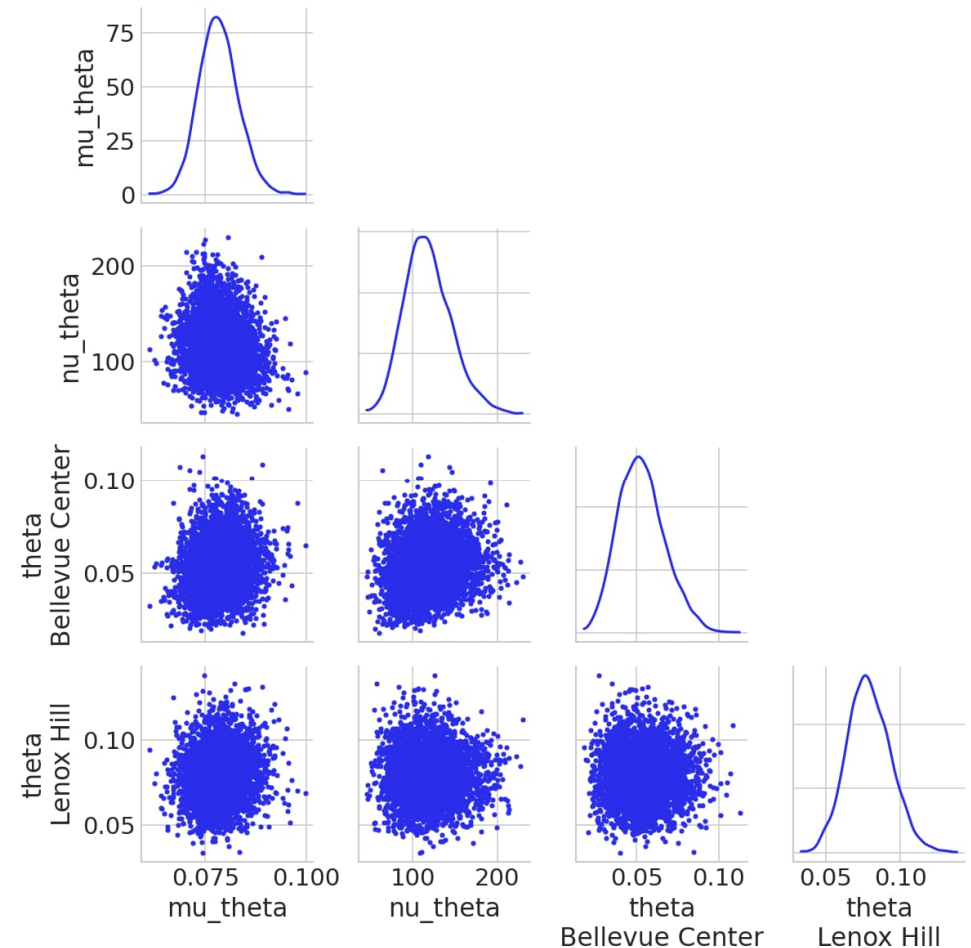
Trace plots for thetas look good

- Only examining a select 5
- No divergences



Pair plots show that parameters are drawn jointly

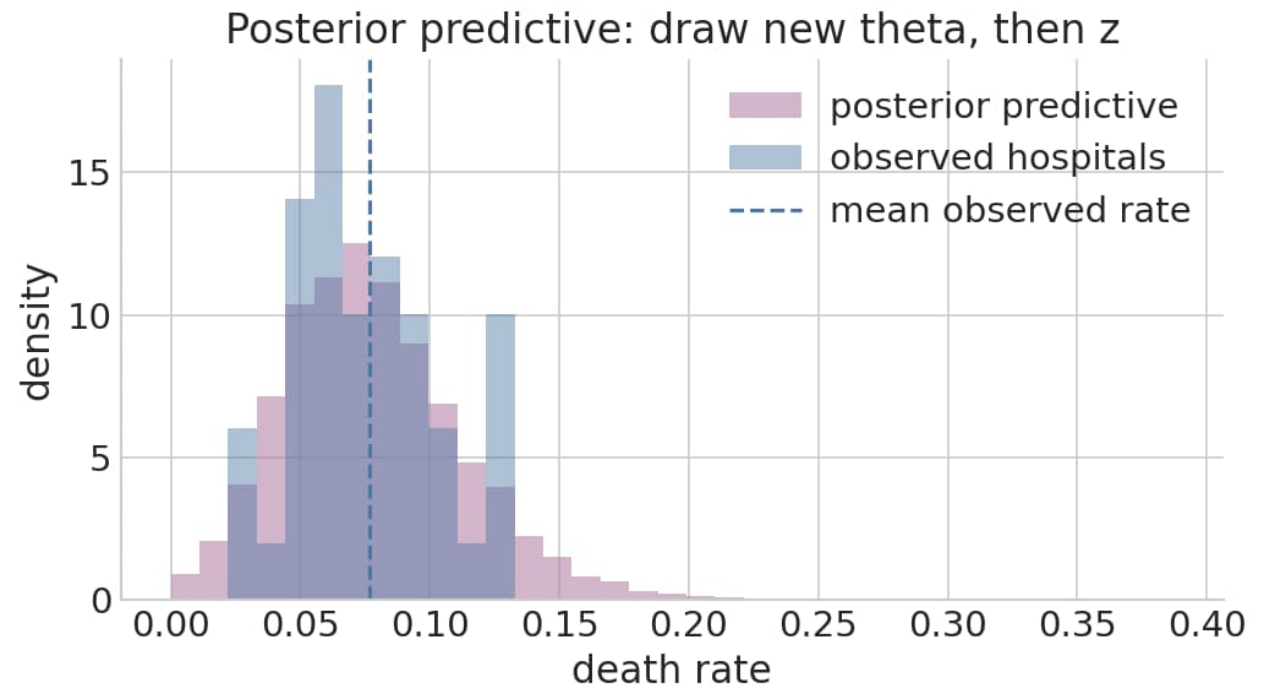
- Every dot is one draw
 - Each draw contains all parameters
 - Parameters are self-consistent
 - Within a draw
- Pair plots are also used for diagnosis
 - Divergences can reveal problematic regions of the posterior
 - Strong correlations may mean the model is not well specified



Posterior predictive data is similar to real data

- For every sample:

- Use sampled μ_θ, ν_θ
- Draw new θ_i
- For each new θ_i
 - Draw a new z_i



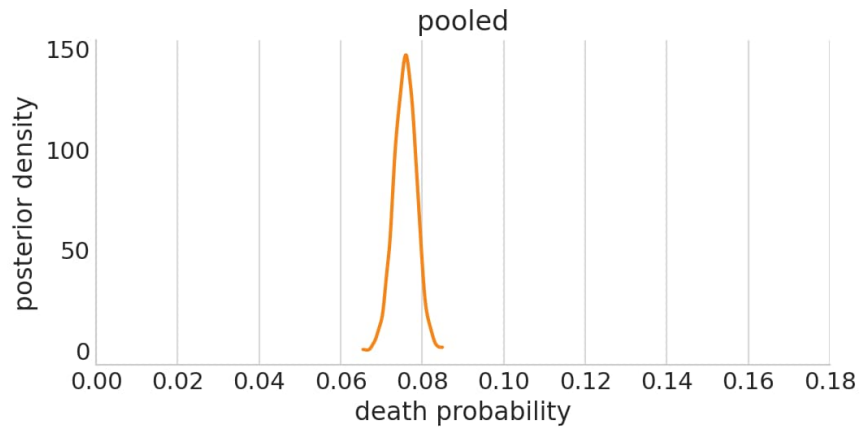
5.7 The posterior distribution

How to explore and interpret the posterior distribution of a complex model

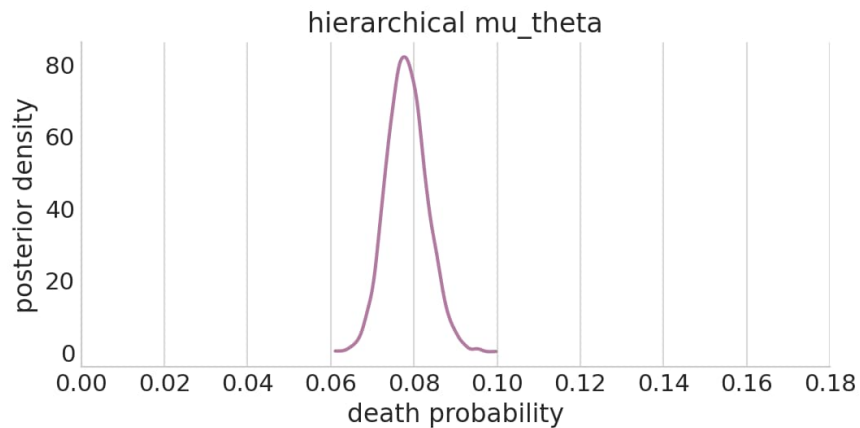
It's now safe to examine posterior

- Prior predictive
 - Shows model is plausible
- Diagnostics
 - Show sampling was successful
- Posterior predictive
 - Shows model fit the data

Hierarchical posterior is different from pooled

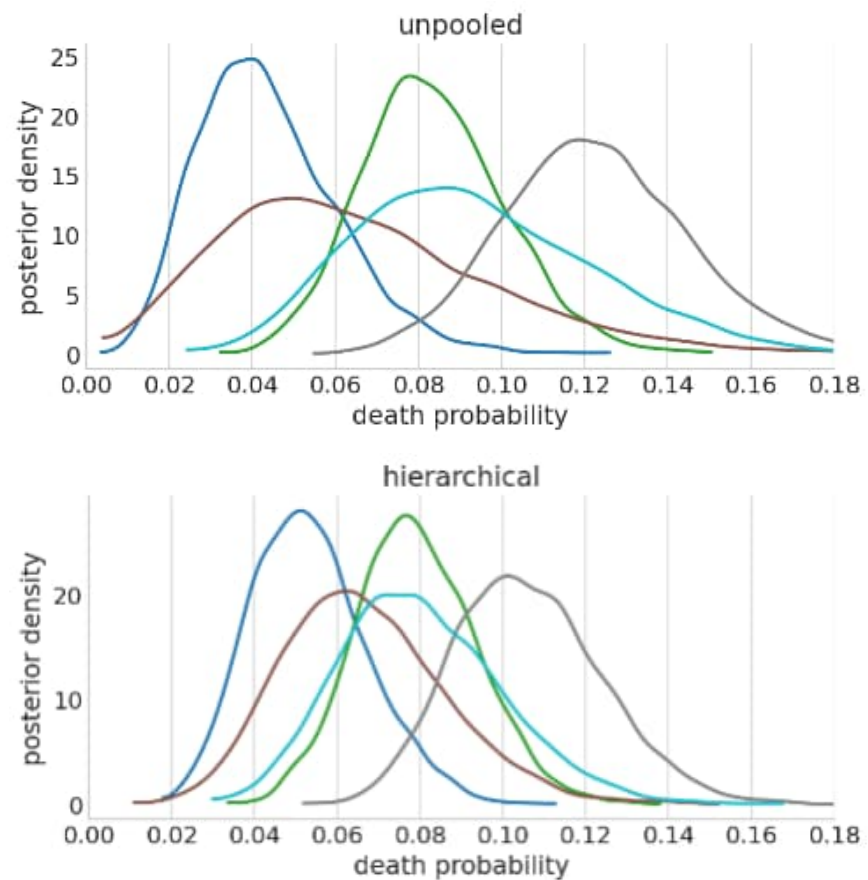


- Less certainty in the mean
 - Reflects mean of 15 hospitals
 - Not mean of 9,235 cases



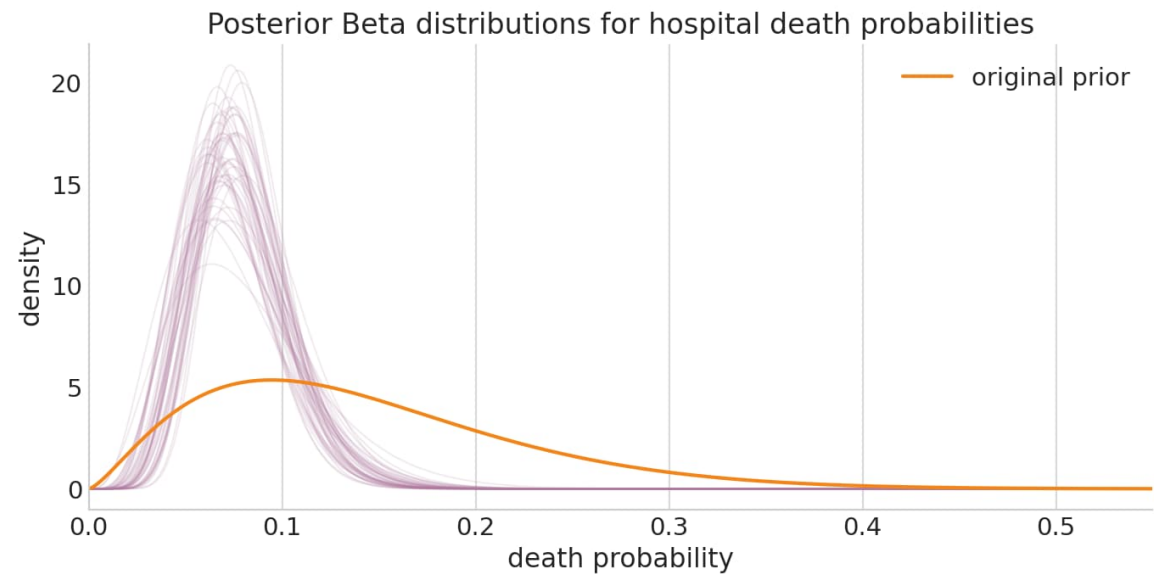
Hierarchical posterior is different from unpooled

- Each hospital more certain
 - Because they inform each other
- Hospitals more similar
 - For the same reason



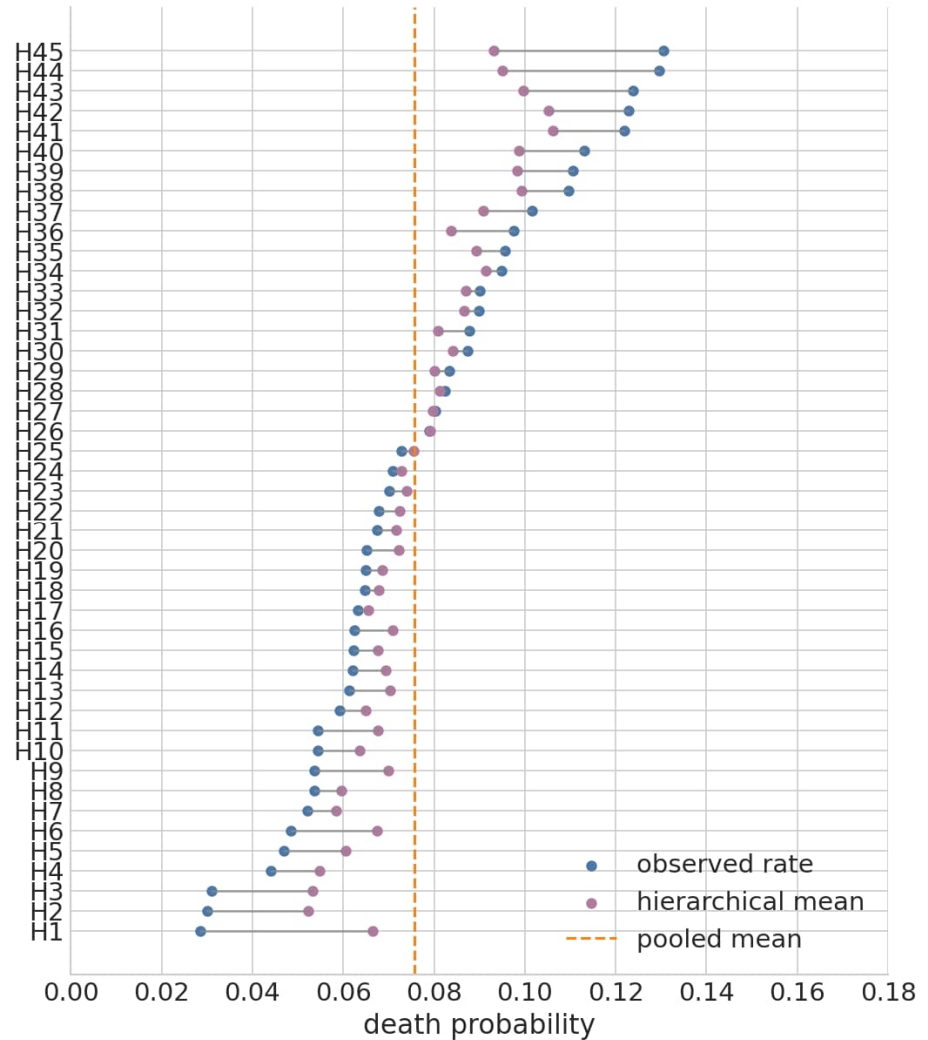
Hyperparameters create a narrower prior

- The prior for θ created by μ_θ, ν_θ is much narrower than the prior we chose for the pooled and unpooled models
 - Reasonable priors are usually broad
 - Hyperparameters use the information in the data to create a prior that reflects the data
- This causes shrinkage



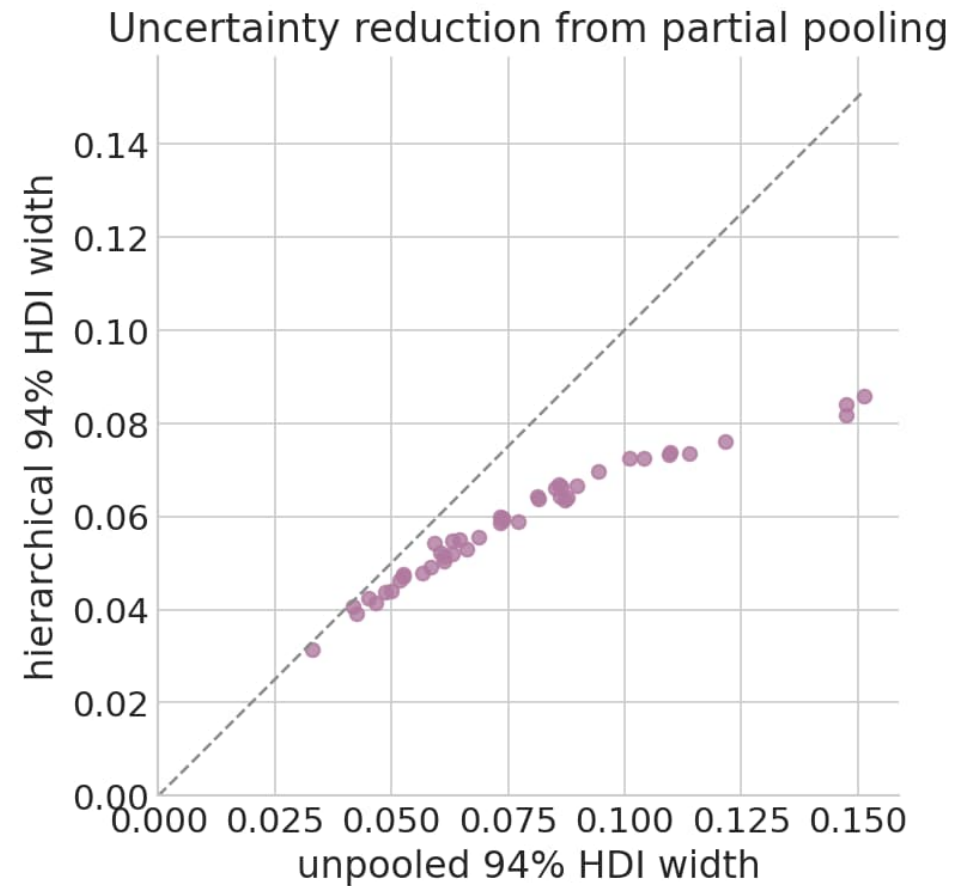
Shrinkage is regression towards the mean

- The estimate of death rate in each hospital is closer to the mean across hospitals than would be expected from the data alone



Hierarchical models also reduce uncertainty

- HDI width of the parameters shrinks
- The effect is stronger where there is less data

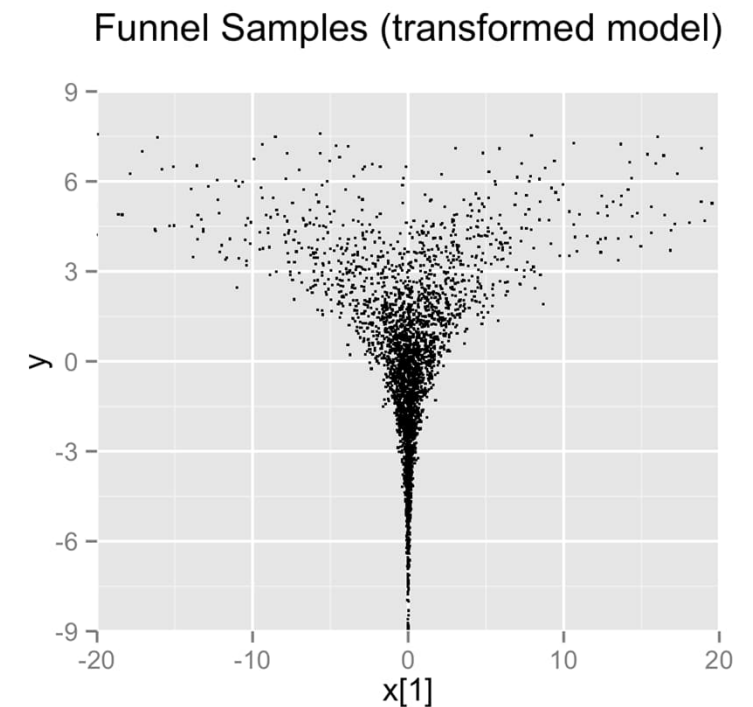


5.8 Funnels

Hierarchical models can create difficult posterior geometry

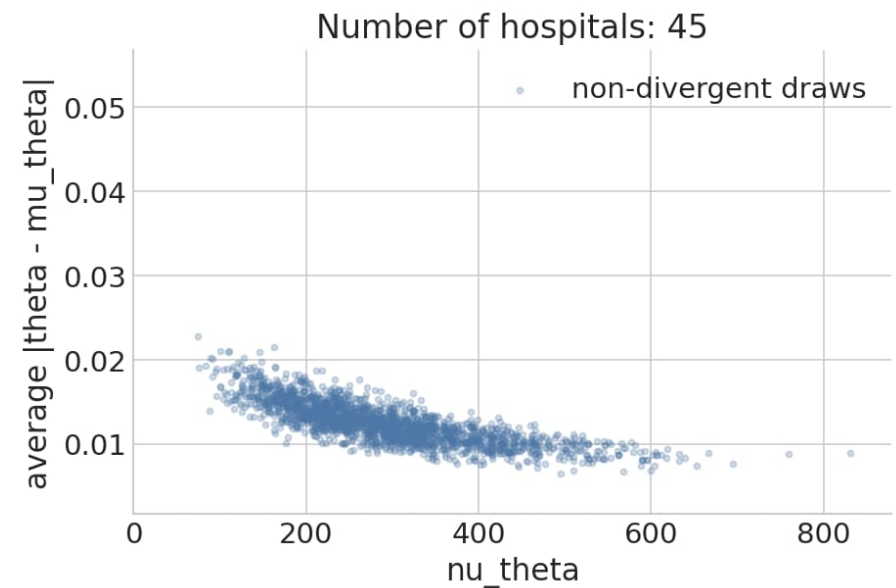
Hierarchical models can create funnels

- Funnels occur when changes in one parameter
 - Usually a width parameter
- Cause changes in scale in other parameters
 - Usually group centers in a hierarchical model
- This geometry is hard for NUTS
 - No step size is right
 - Either too small or too big



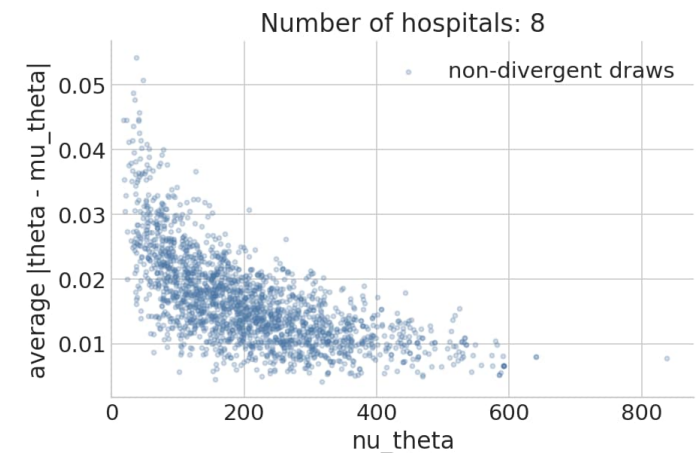
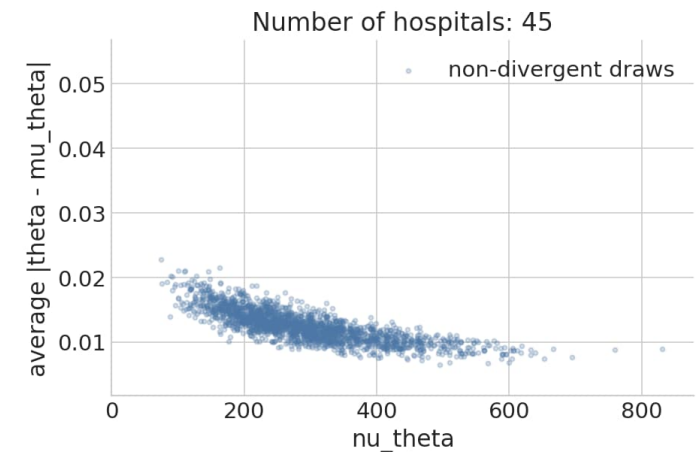
Our hierarchical model has a funnel

- As ν_θ gets large
 - θ get closer and closer together



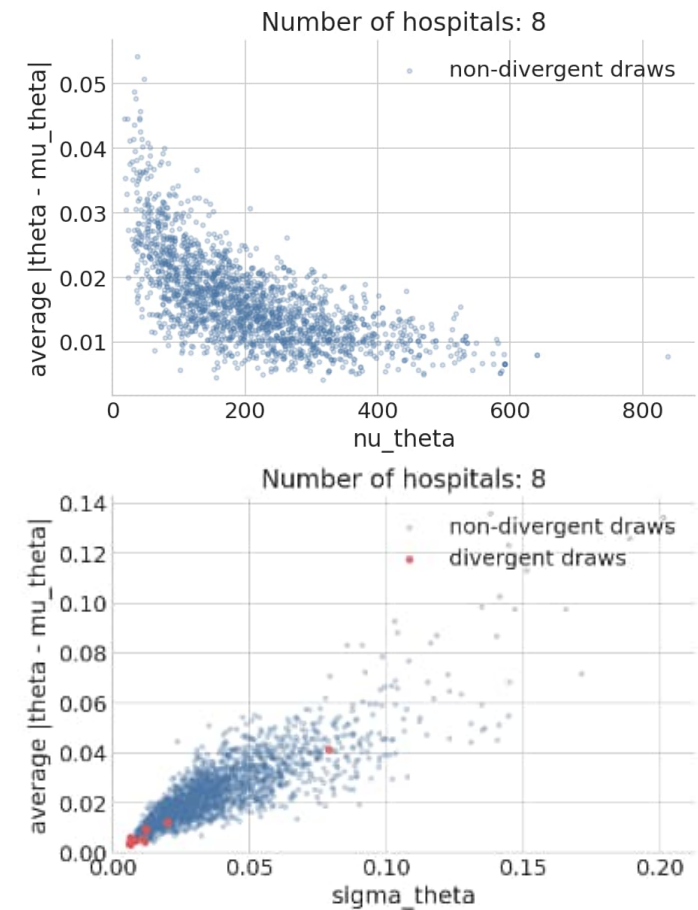
Parameterization affects funnel behavior

- The larger dataset succeeds in exploring the parameter space more fully
- Biased estimate caused by the funnel
 - Explore part of the space more easily



Funnels cause more trouble in smaller datasets

- Changing the beta parameterization from width to precision makes the funnel problem worse
 - There is less 'room' when width is close to 0
 - There is more 'room' when concentration is close to infinity



Location and scale parameters

- Location parameters
 - Measure where the center of the distribution is
 - Mean, median, mode and others
- Scale parameters
 - Measure how broad the center of the distribution is
 - Standard deviation, variance, interquartile range, and others

When to worry about funnels

- An unknown scale is used as the scale of an unknown location
- The scale parameter can go to 0
 - The data is weak
 - The prior allows 0

5.9 Multi-level models

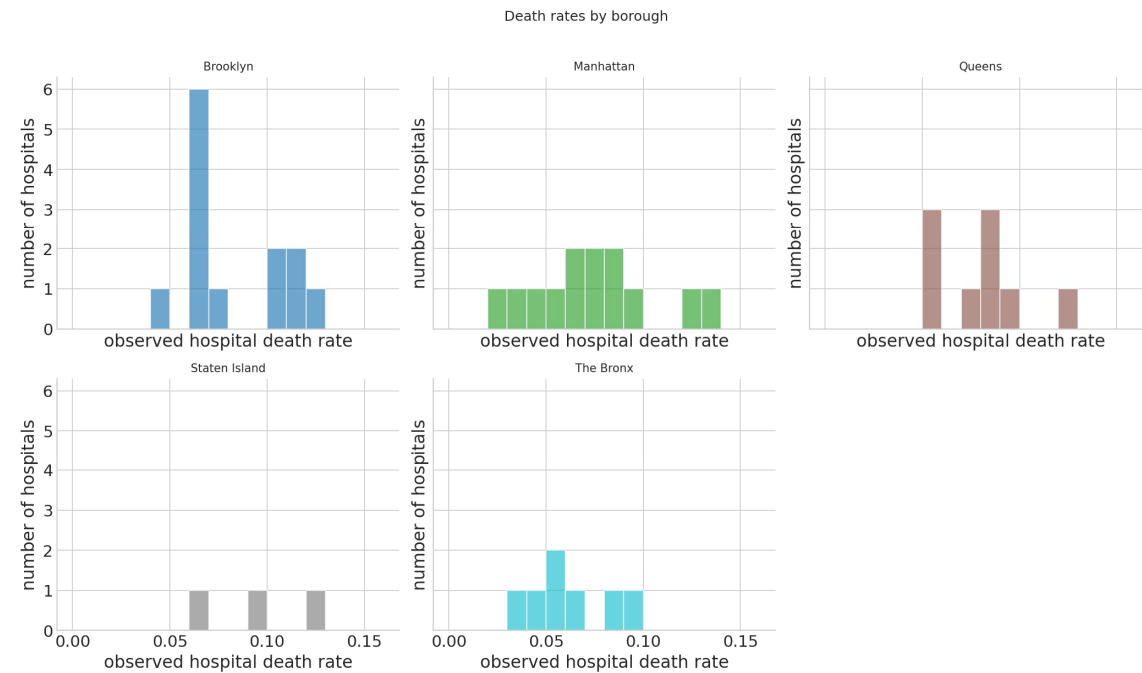
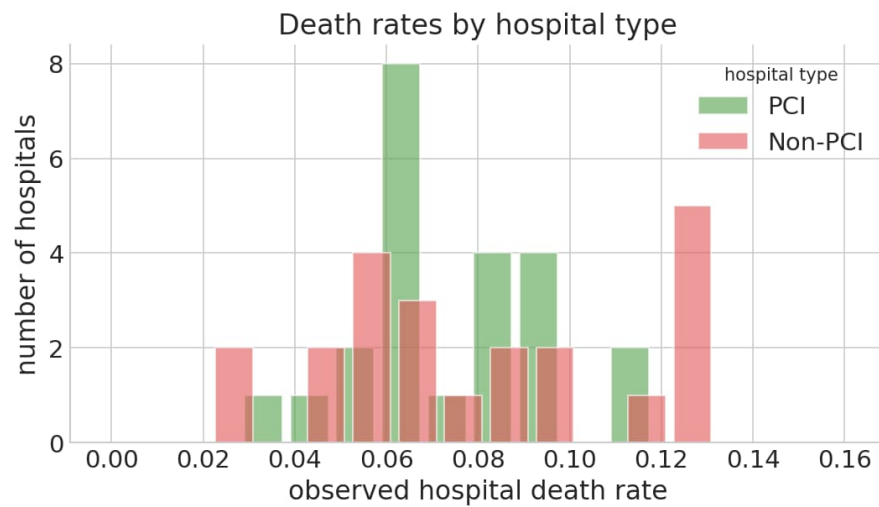
Bayesian models can have multiple levels of hierarchy

Hierarchy can come at multiple levels

- MRI scans
 - Taken on 100 different individual machines
 - That are all from 5 different models
 - And the models are all from a single population
- Firing rates of neurons
 - From different brain areas
 - In different individuals
 - And the overall population of neurons
- Tests of biomaterial strengths
 - Generated in multiple batches
 - Using different formulations
 - And the overall properties of the material

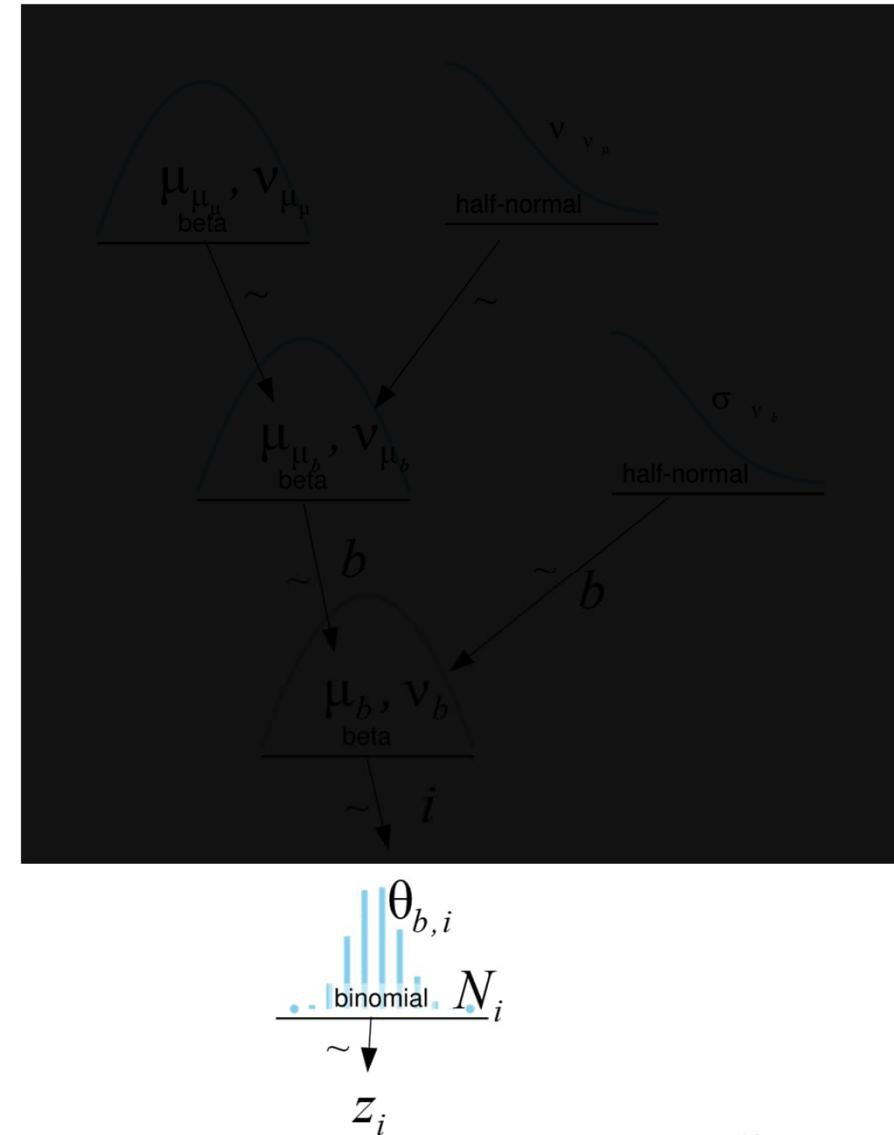
The hospital data has different groupings

- Two different hospital types
 - With different cardiac arrest technologies
- Five different boroughs
 - With different populations and incomes



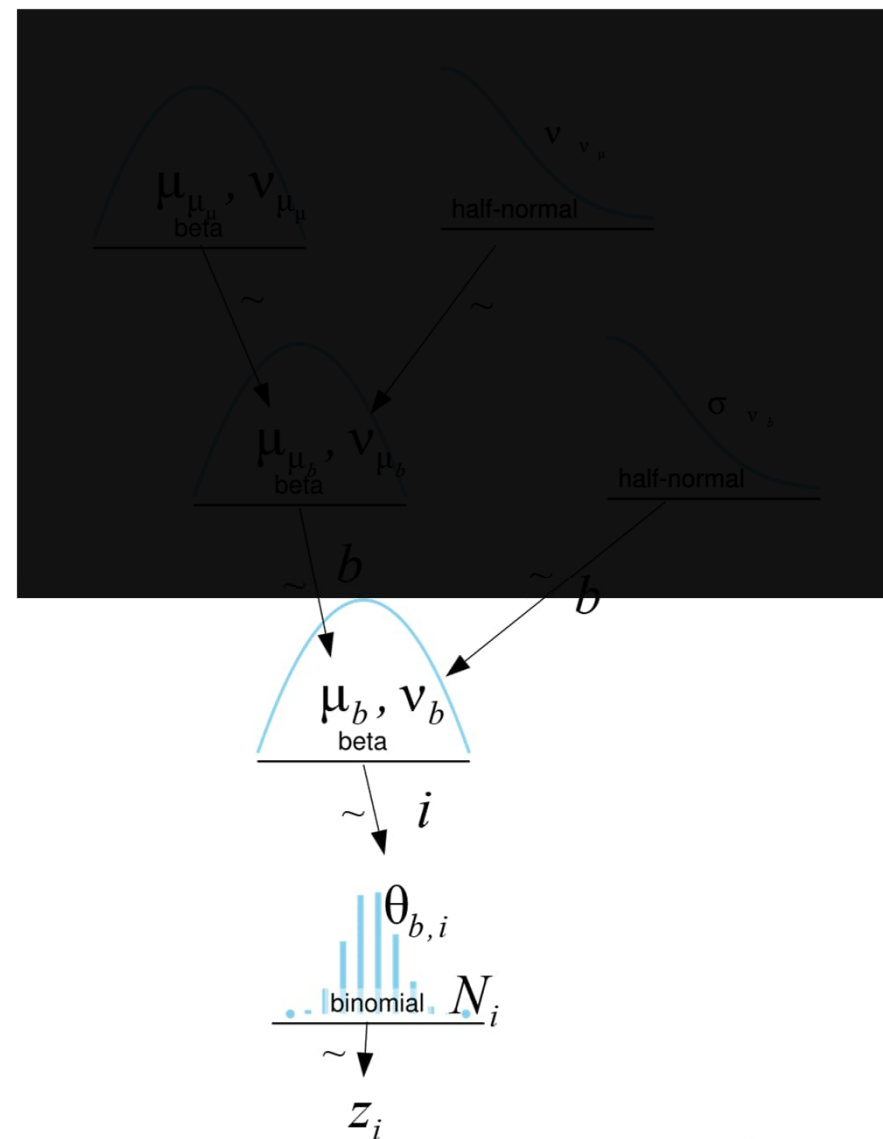
Model step 1: likelihood

- Each hospital
 - Probability parameter
 - Number of cases
 - Number of deaths



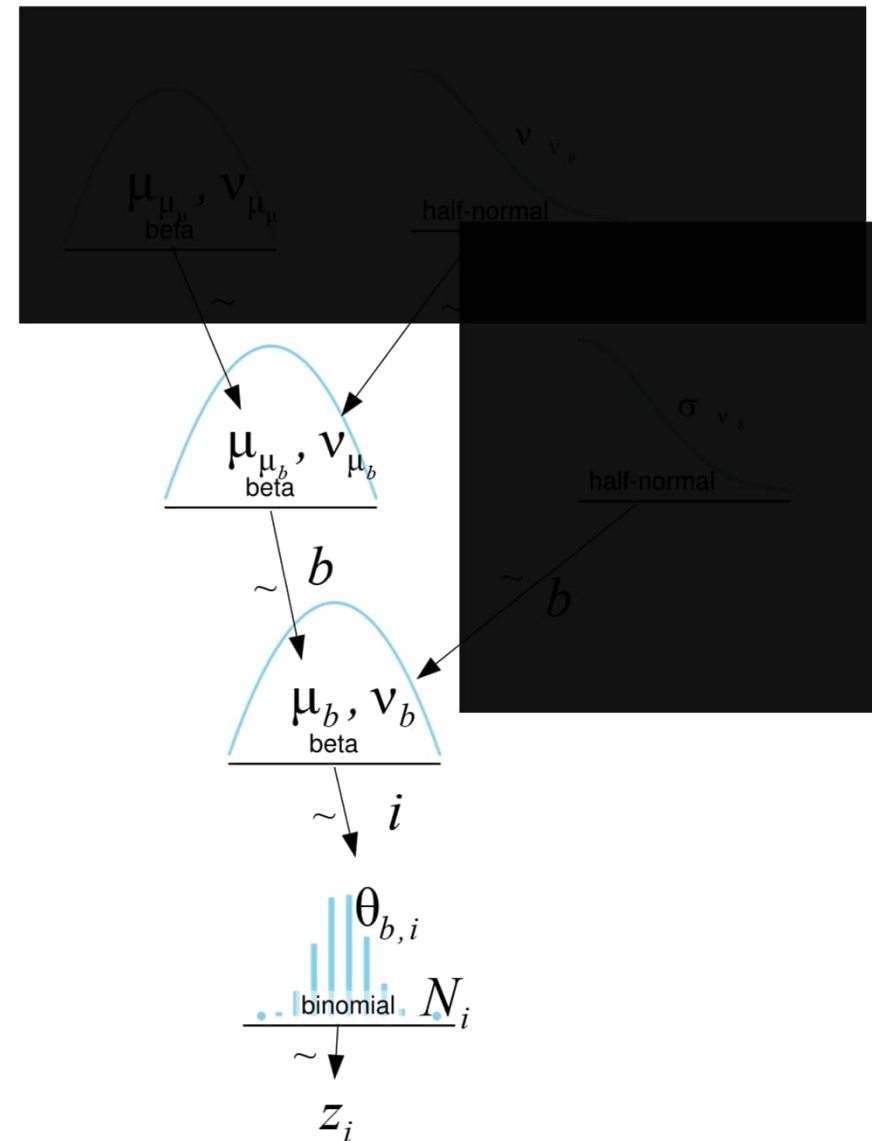
Model step 2: parameter

- Each hospital
 - Deaths and rate
- Each borough:
 - Probability distribution
 - Mean death rate: μ_b
 - Similarity within borough: ν_b



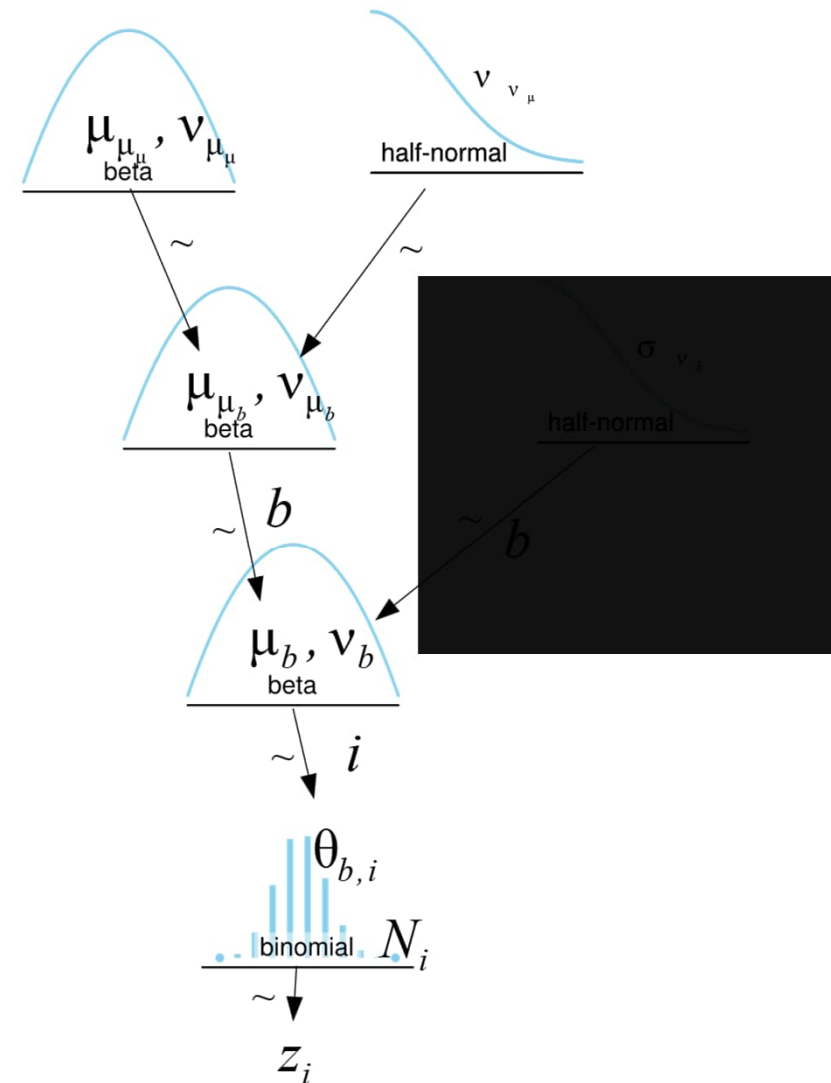
Model step 3: hyperparameter

- Each hospital
 - Deaths and rate
- Each borough
 - Distribution of rates
- Across boroughs:
 - Distribution of mean rates
 - Mean across boroughs: μ_{μ_b}
 - Similarity of borough means: ν_{μ_b}



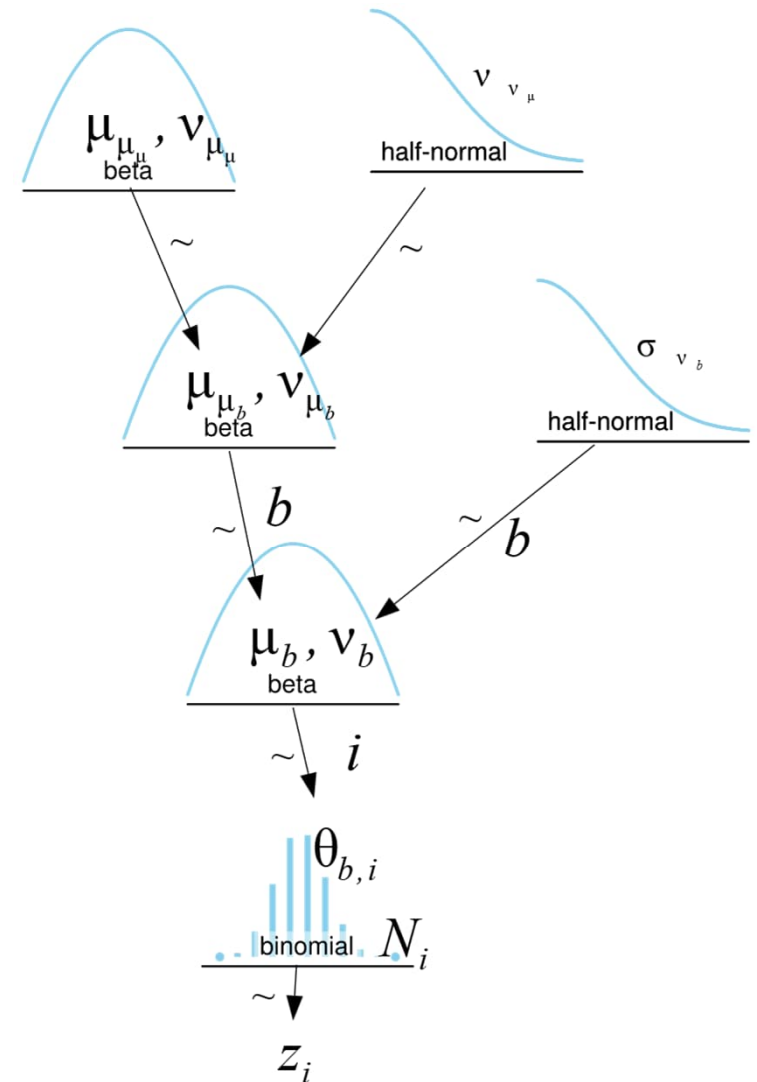
Model step 4a: hyperprior

- Each hospital
 - Deaths and rate
- Each borough
 - Distribution of rates
- Across boroughs:
 - Distribution of mean rates
 - Mean across boroughs: μ_{μ_b}
 - Similarity of borough means: ν_{μ_b}
 - Hyperpriors for both of these



Model step 4b: hyperprior 2

- Each hospital
 - Deaths and rate
- Each borough
 - Distribution of rates
- Across boroughs:
 - Distribution of mean rates
 - Mean across boroughs: μ_{μ_b}
 - Similarity of borough means: v_{μ_b}
 - Hyperpriors for both of these
 - Distribution of similarity within borough
 - Hyperprior on this



Model code

```
with pm.Model(coords=coords_group) as group_beta_binomial_model:
```

```
    mu_mu_theta = pm.Beta("mu_mu_theta", mu=omega_mu, nu=omega_nu)
```

```
    nu_mu_theta = pm.HalfNormal("nu_mu_theta", sigma=20)
```

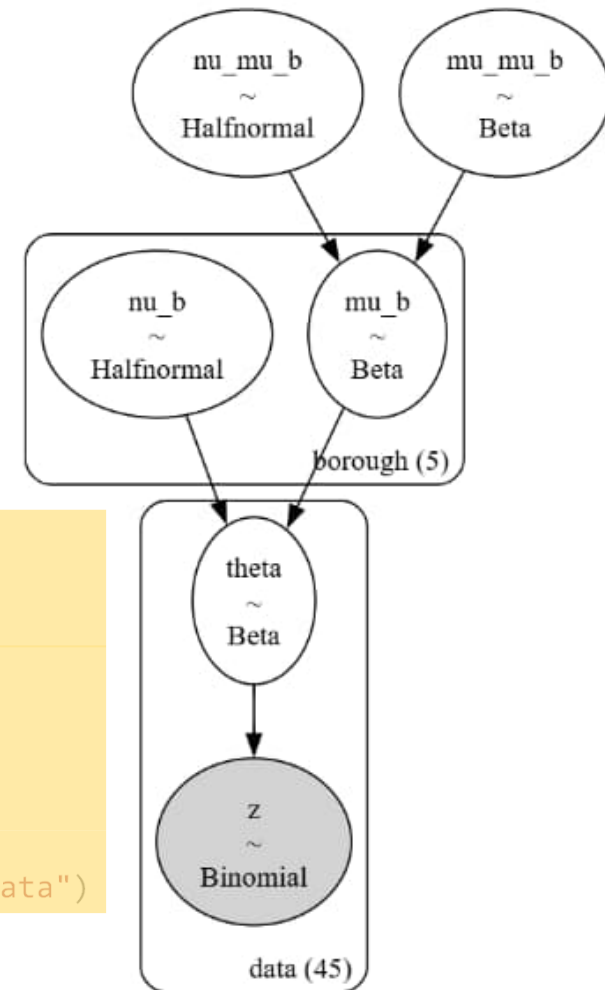
```
    sigma_nu_theta = pm.HalfNormal("sigma_nu_theta", sigma=20)
```

```
    mu_theta = pm.Beta("mu_theta", mu=mu_mu_theta, nu=nu_mu_theta, dims="group")
```

```
    nu_theta = pm.HalfNormal("nu_theta", sigma=sigma_nu_theta, dims="group")
```

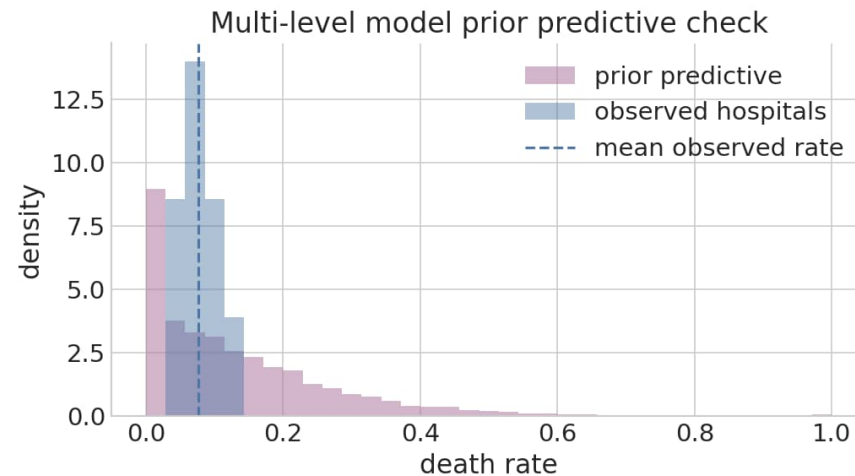
```
    theta = pm.Beta("theta", mu=mu_theta[group_idx], nu=nu_theta[group_idx], dims="data")
```

```
    z = pm.Binomial("z", n=cases, p=theta, observed=deaths, dims="data")
```



Setting priors

- $\mu_{\mu\mu}$ and $\nu_{\mu\mu}$
 - Parameters of the grand mean
 - Can use the same prior as earlier hierarchical model
- σ_{ν_b}
 - Scale of similarity of hospitals within boroughs
 - Can use the same prior as earlier similarity of hospitals
- σ_{ν_μ}
 - Scale of similarity of boroughs
 - Probably safe to re-use σ_{ν_b}



The parameters in the posterior

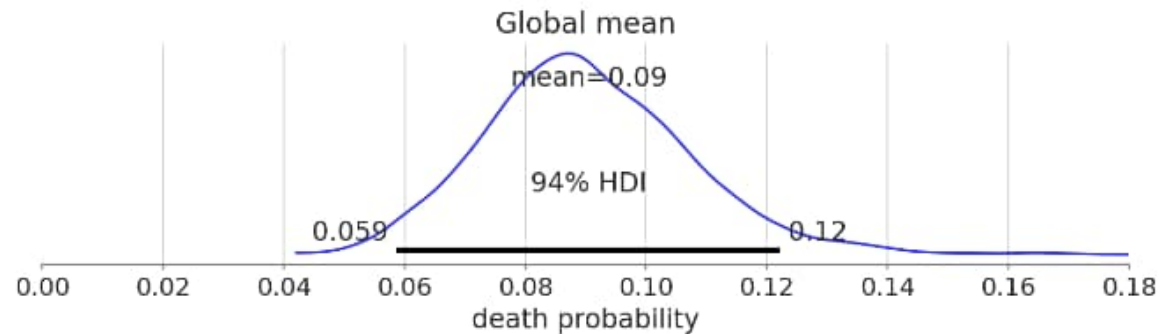
	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
mu_mu_b	0.090	0.018	0.059	0.122	0.000	0.000	4846.0	2768.0	1.0
nu_mu_b	70.373	30.562	14.670	124.468	0.401	0.509	5301.0	2632.0	1.0
mu_b[Manhattan]	0.076	0.011	0.056	0.096	0.000	0.000	5708.0	3304.0	1.0
mu_b[Brooklyn]	0.085	0.011	0.067	0.105	0.000	0.000	6224.0	3135.0	1.0
nu_b[Manhattan]	78.144	28.460	29.164	131.842	0.409	0.399	4470.0	2869.0	1.0
nu_b[Brooklyn]	81.017	27.708	32.824	132.528	0.359	0.373	5563.0	2948.0	1.0
theta[Harlem Center]	0.060	0.025	0.017	0.108	0.000	0.000	6294.0	2360.0	1.0
theta[Bellevue Center]	0.047	0.016	0.020	0.077	0.000	0.000	7299.0	2874.0	1.0
theta[Wyckoff Heights MC]	0.060	0.016	0.032	0.090	0.000	0.000	7344.0	2748.0	1.0

The mean across boroughs

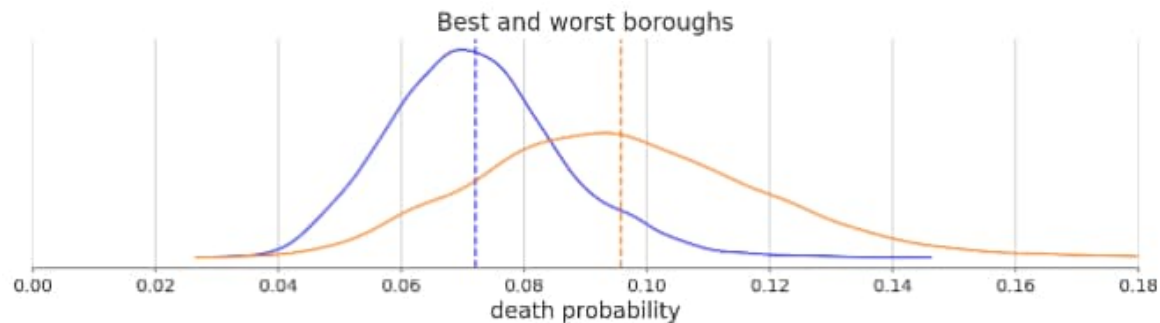
The mean of each borough

The death rates of hospitals

Comparing population, position and individual means



Death rate is around 9%



The Bronx is around 7%
Staten Island around 9.5%
But there is less data on
Staten Island



Each borough has a range of
hospitals

5.10 Summing up / looking forward

Benefits and pitfalls of Bayesian models + where we go next time

Benefits and pitfalls of hierarchical models

Benefits

- Model structured data
- Balance
 - Population information
 - Group information
- Robust estimation
 - Even if some groups have less data

Pitfalls

- Computational complexity
 - Increased fitting time
 - Importance of sampling diagnostics
- Priors disconnected from data
 - Importance of prior predictive
- Interpretational complexity
 - Simultaneous interpretation at multiple levels

Next time!

- We finally learn what it means for something to be “statistically significant”
 - How to understand and interpret differences between groups